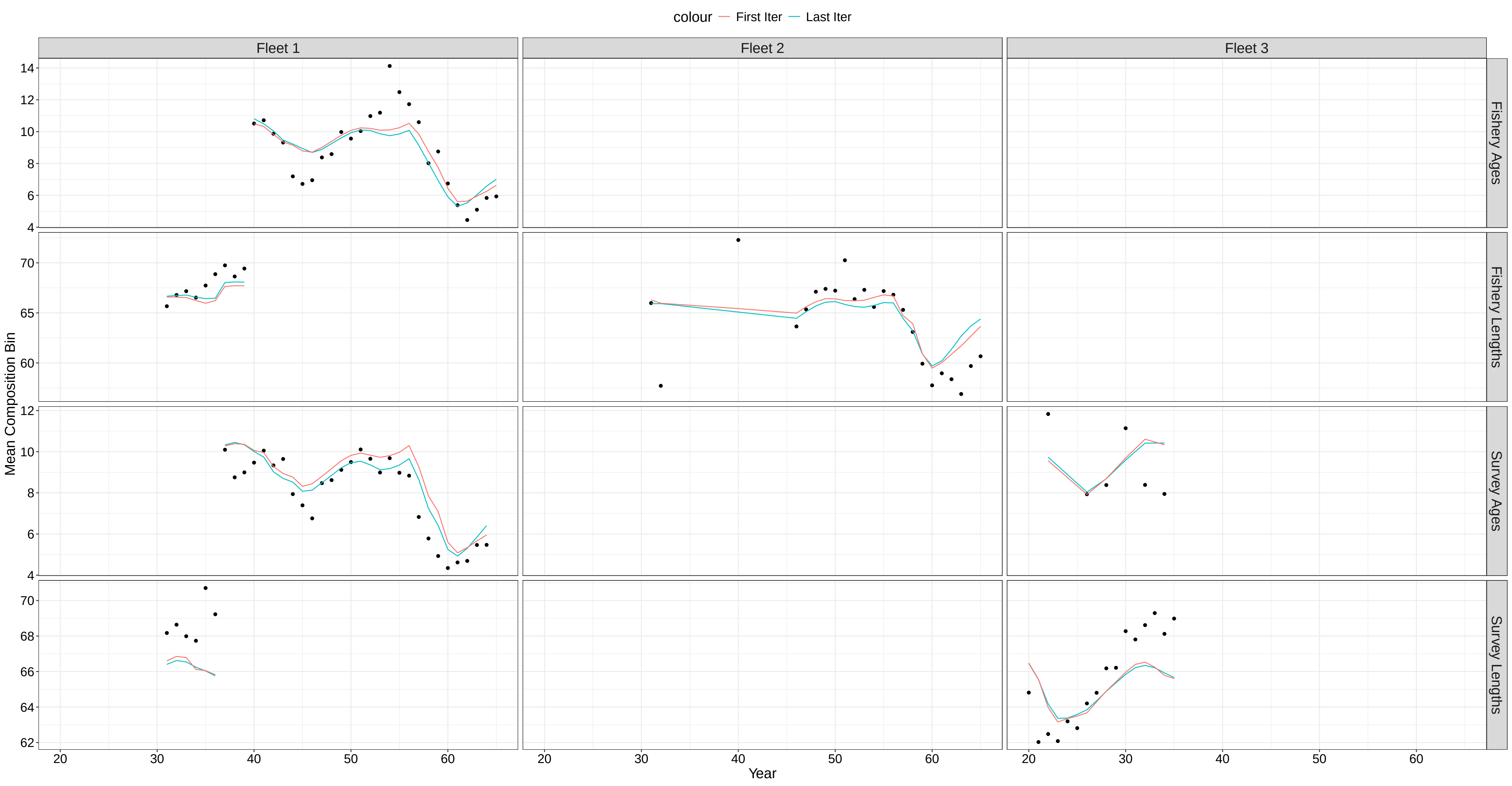
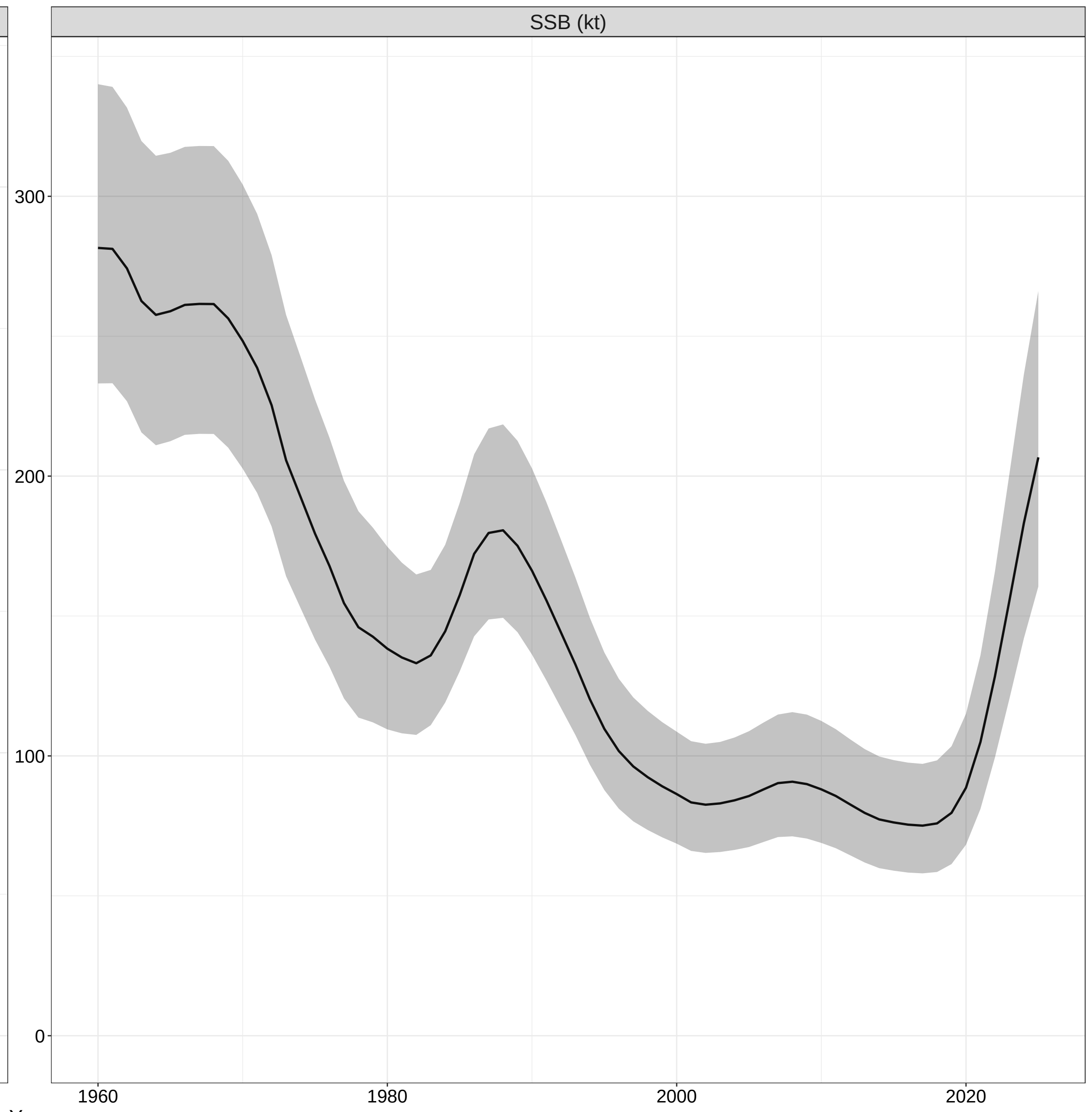
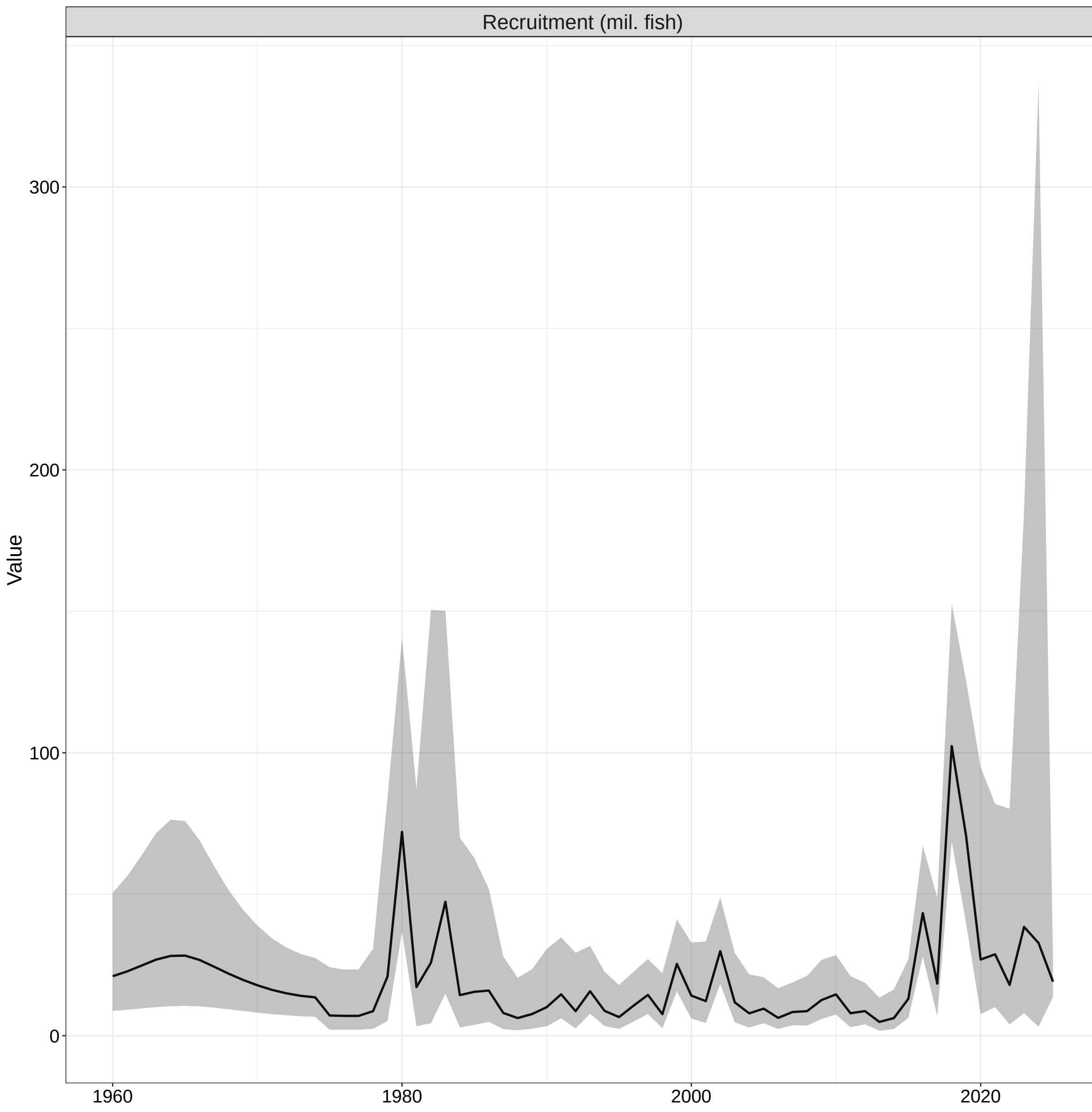


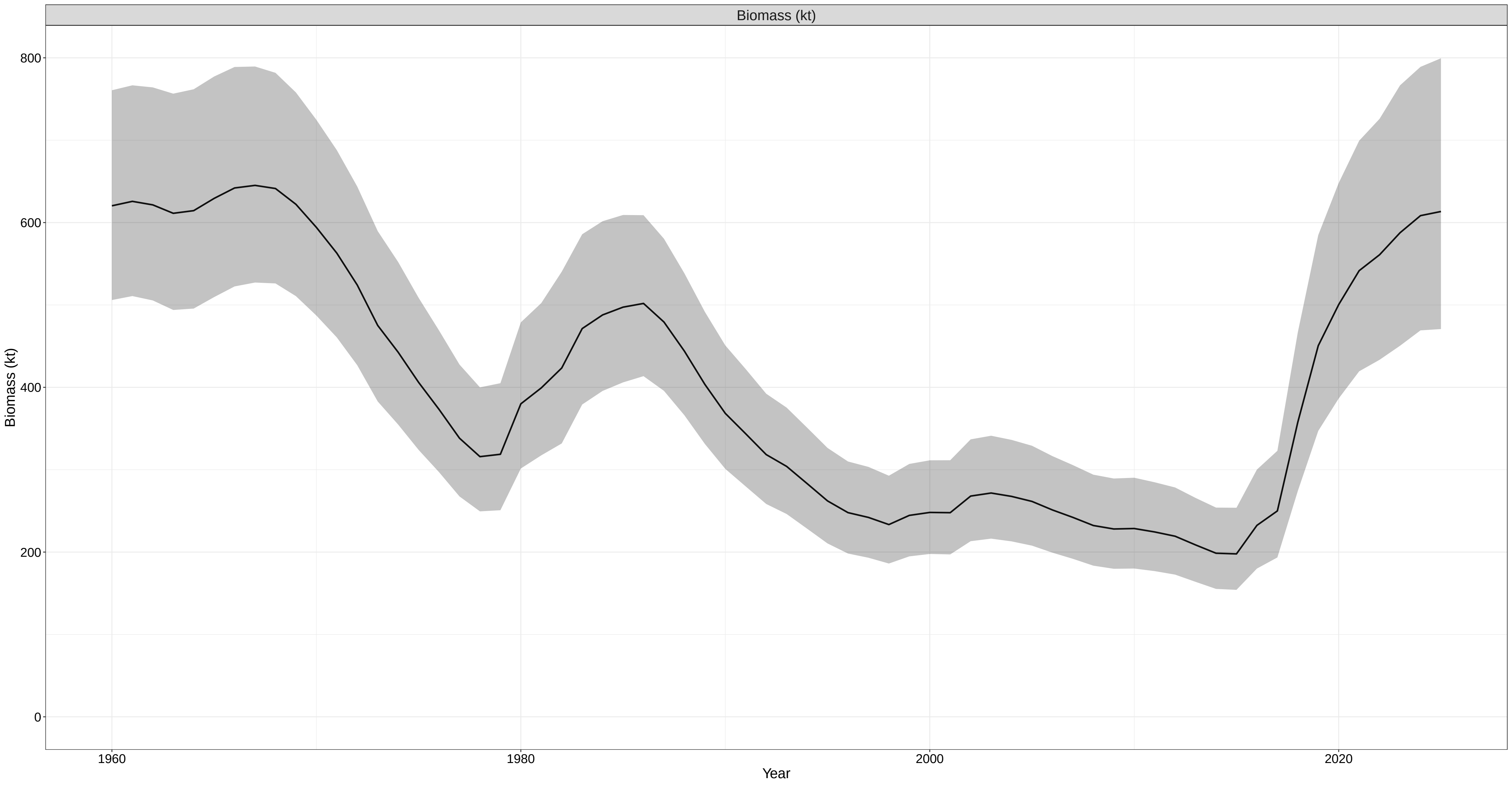
Source	SE	Eff_SE	Data_Wt	ISS	ESS
LL Survey Index	0.09	0.09	1.00	NA	NA
JPN LL Survey Index	0.09	0.09	1.00	NA	NA
Fixed Gear Fishery CPUE	0.20	0.20	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.80	20	16.00
LL Survey Ages	NA	NA	3.54	20	70.75
JPN LL Survey Ages	NA	NA	0.64	20	12.87
Fixed Gear Fish Lengths	NA	NA	3.48	20	69.54
Trawl Fish Lengths	NA	NA	0.35	20	6.99
LL Survey Lengths	NA	NA	2.29	20	45.75
JPN LL Survey Lengths	NA	NA	1.03	20	20.56

37504e−05 which was greater than tolerance 1e−05. This indicates potential non-convergence according to the tolerance.



	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	206.69993	357.02075	0.04784667963714	39.55138	131.45493	0.07875	1.5724	0.57896	0.60758

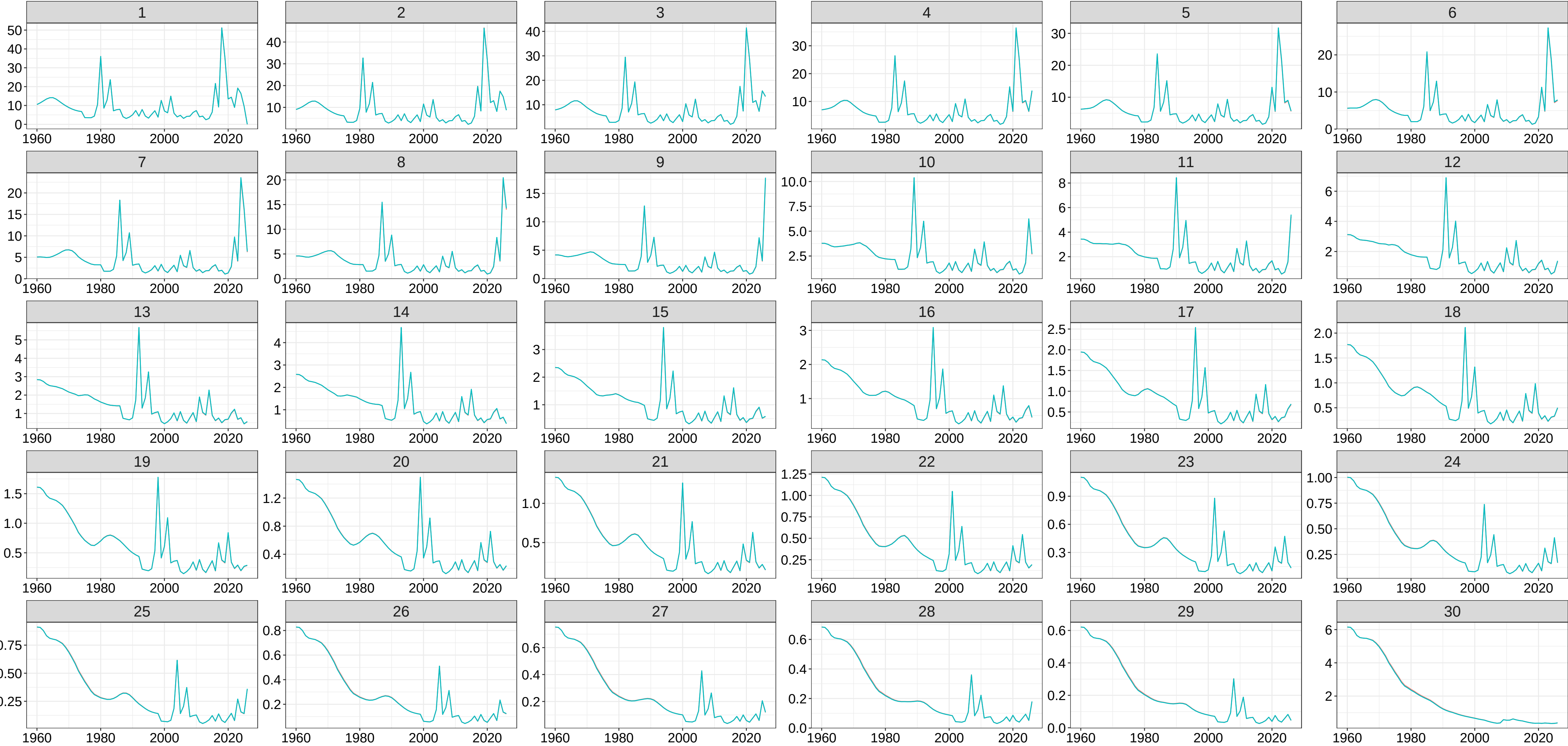




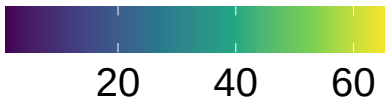
Numbers-at-Age

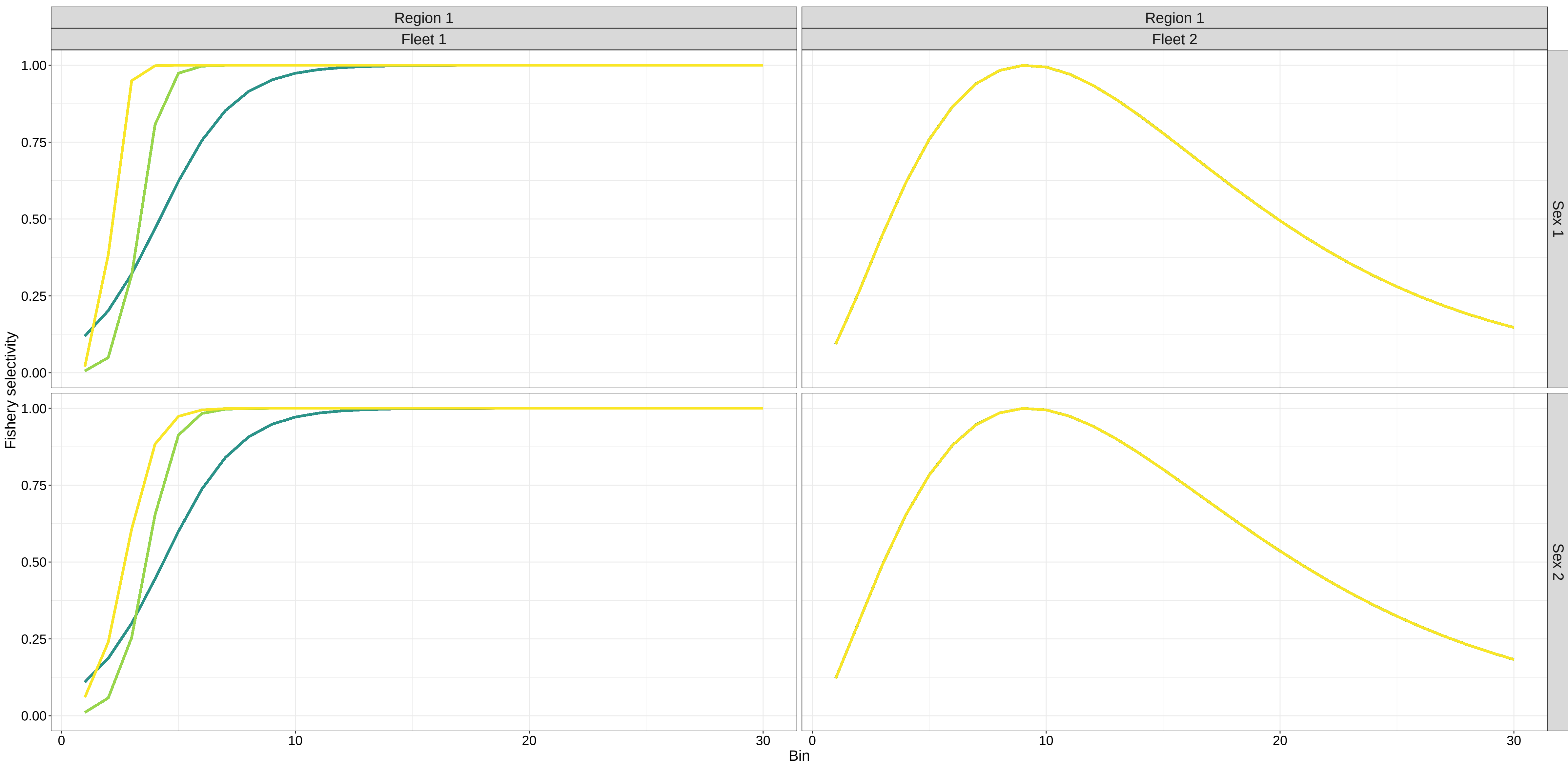
Sex — Female — Male

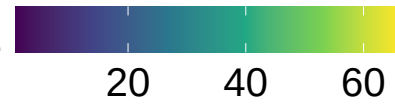
Numbers-at-age (millions)

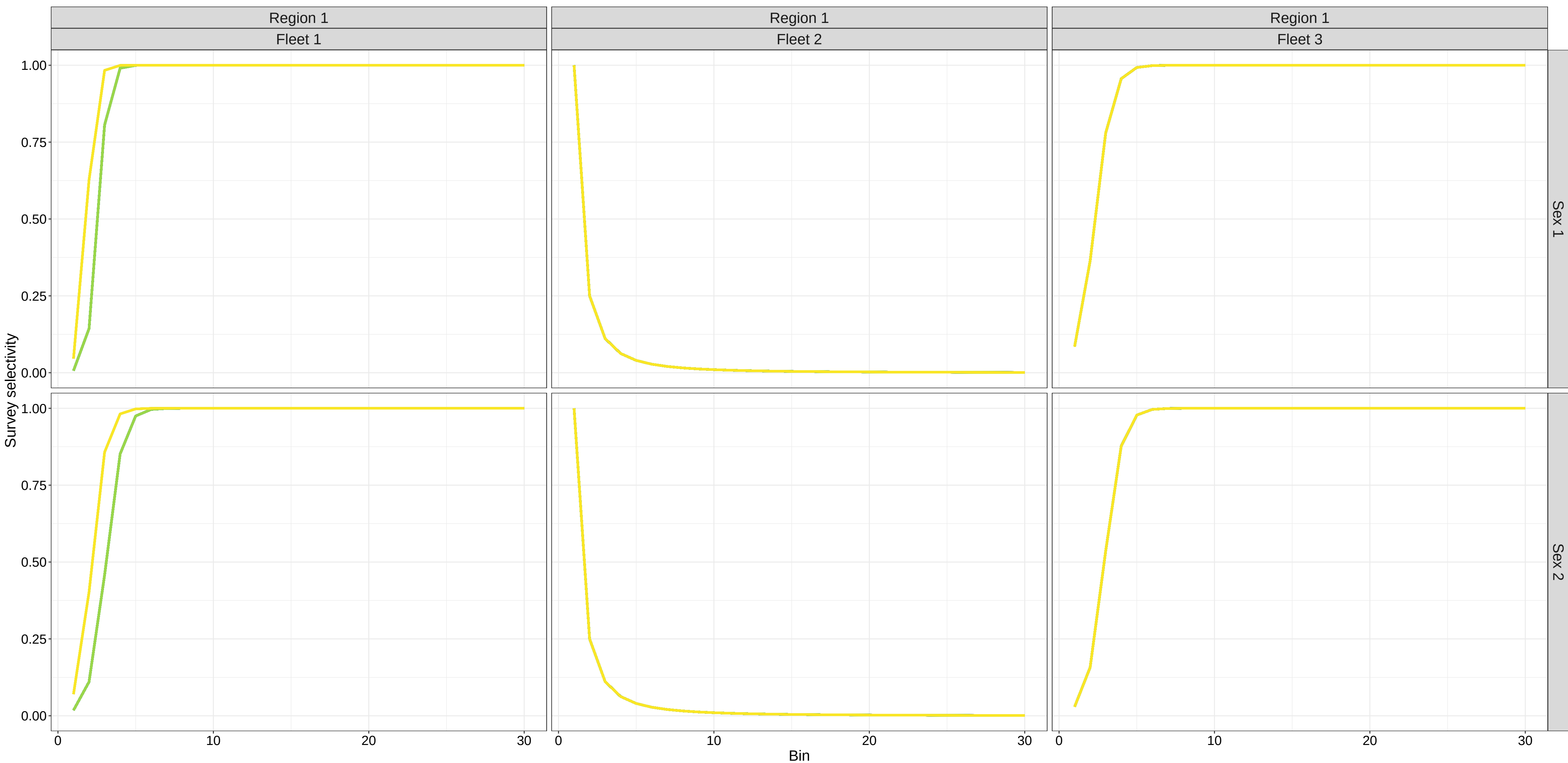


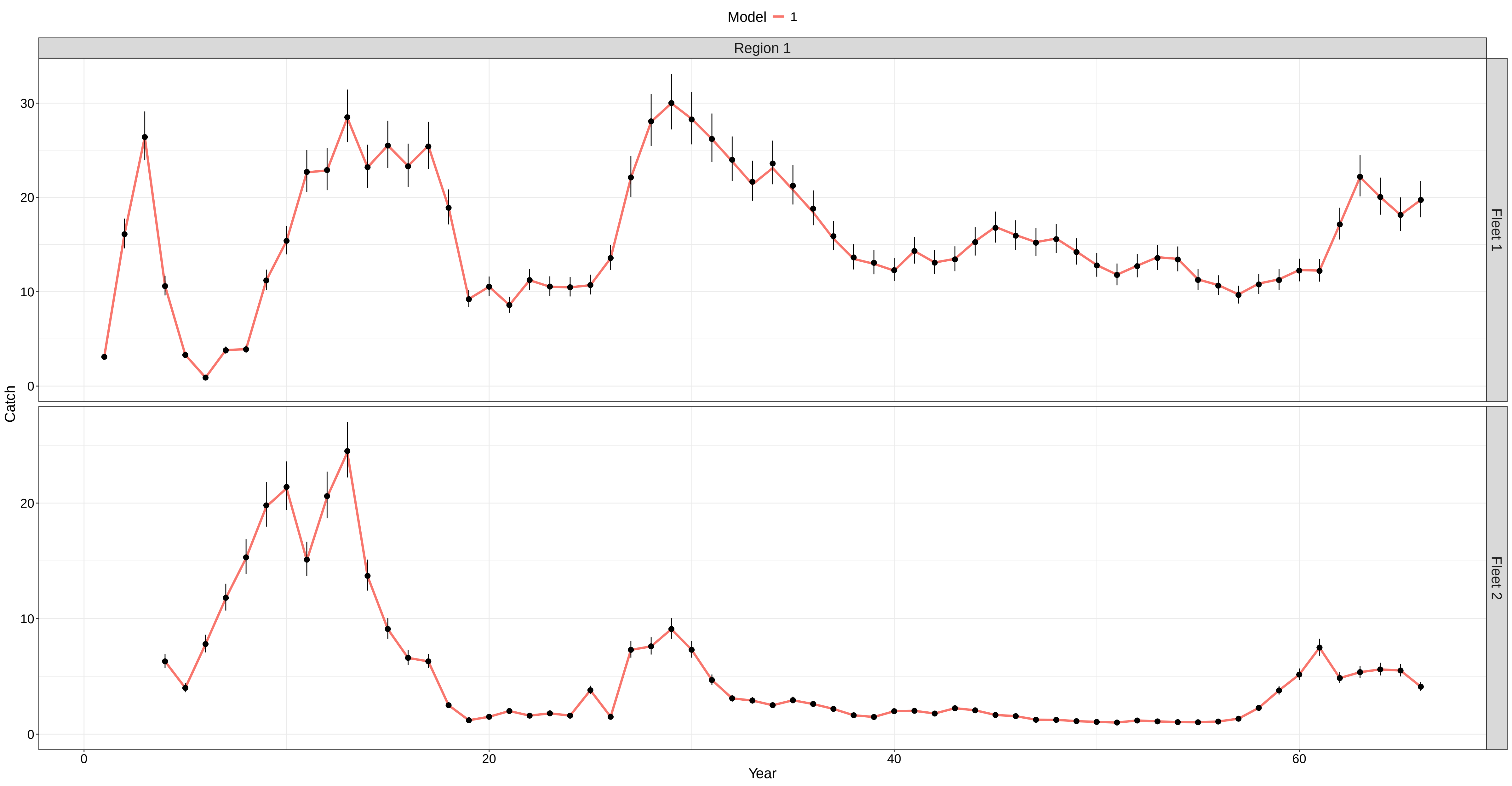
Year

Model — 1 Year 

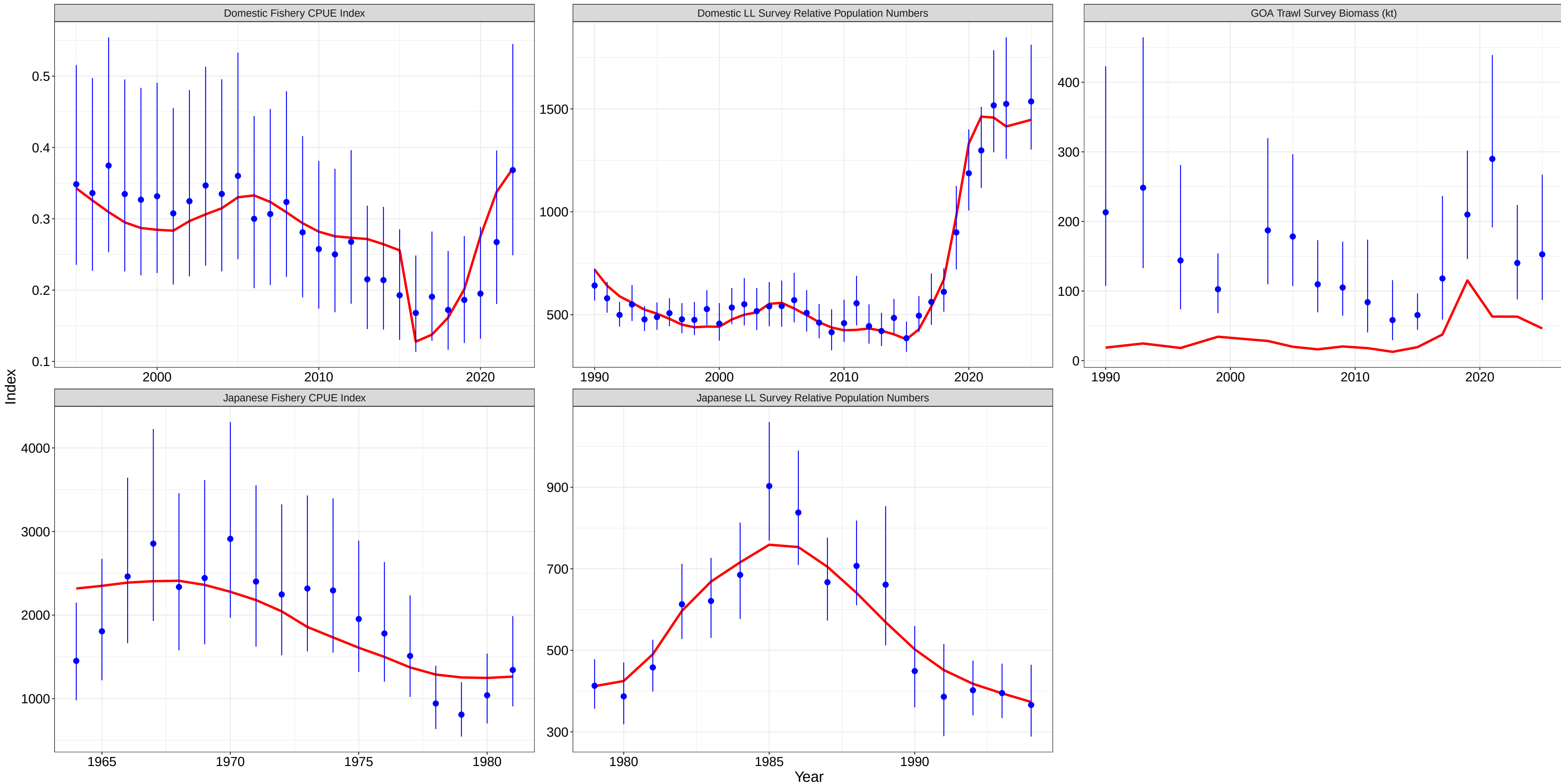


Model — 1 Year 

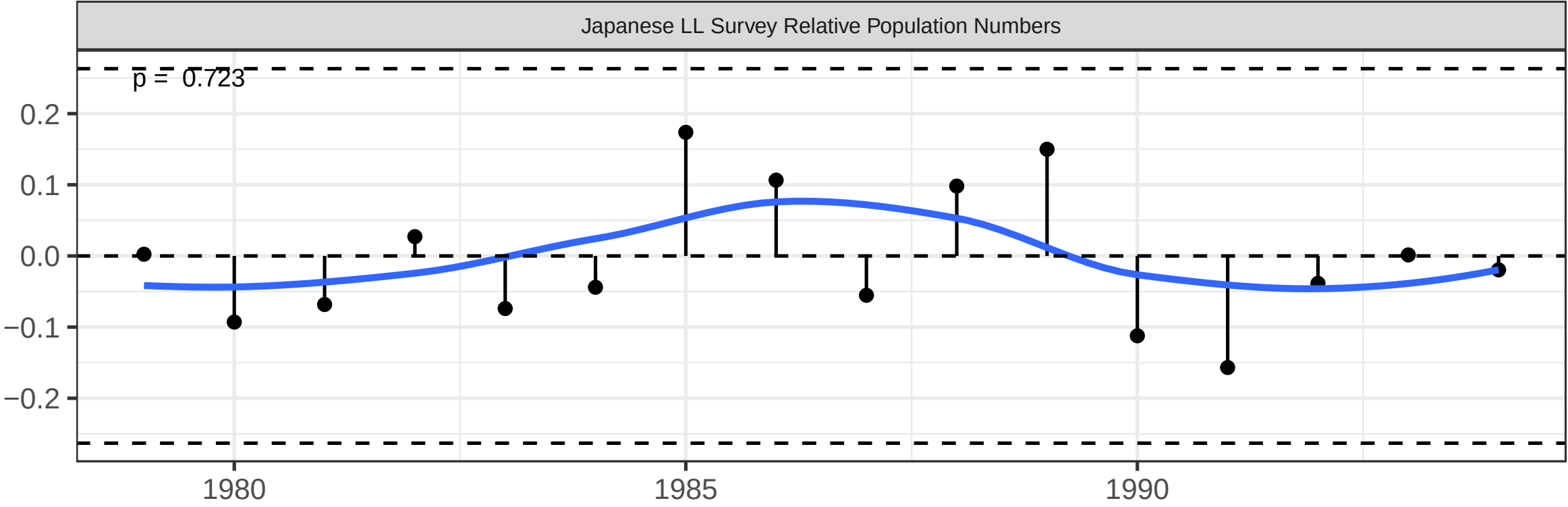
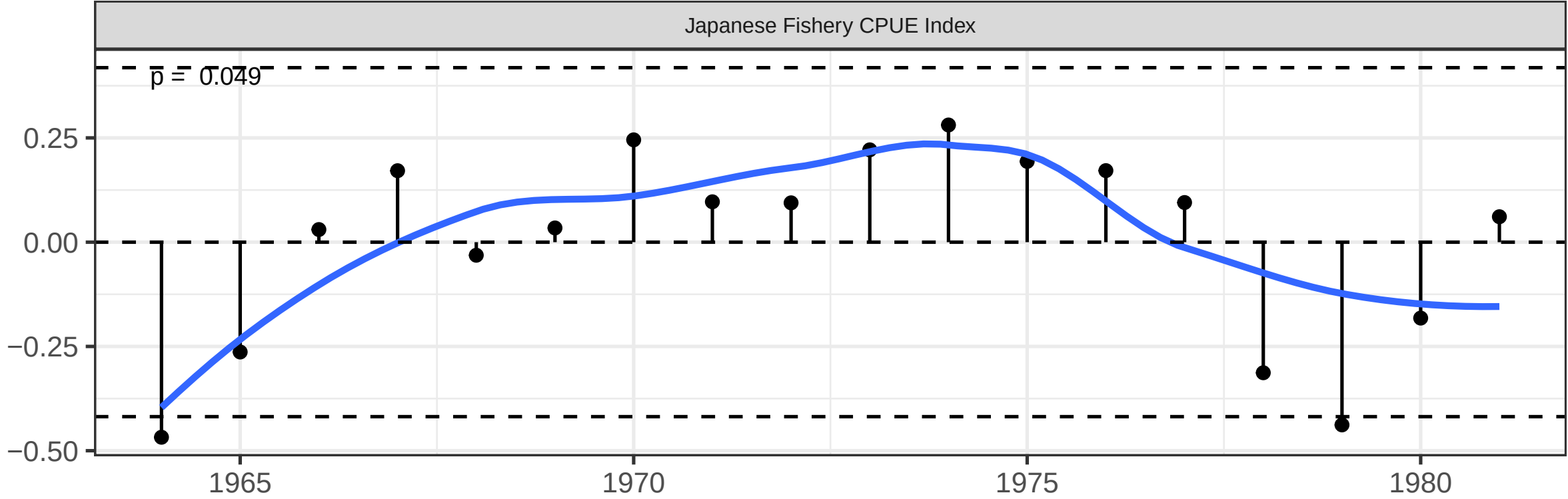
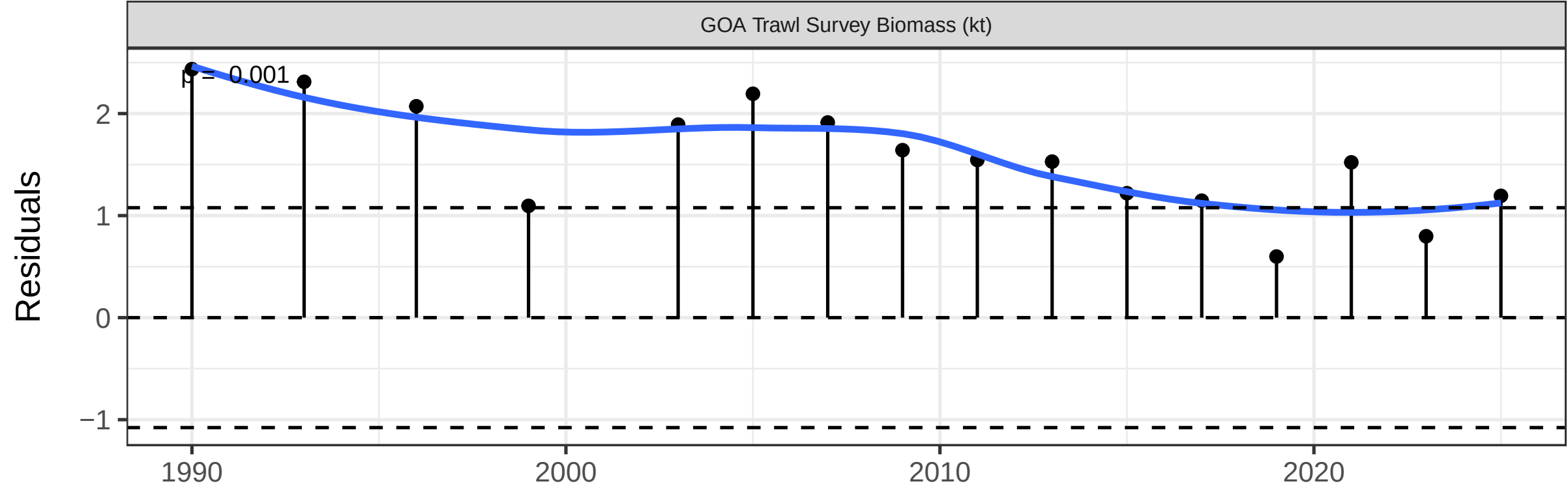
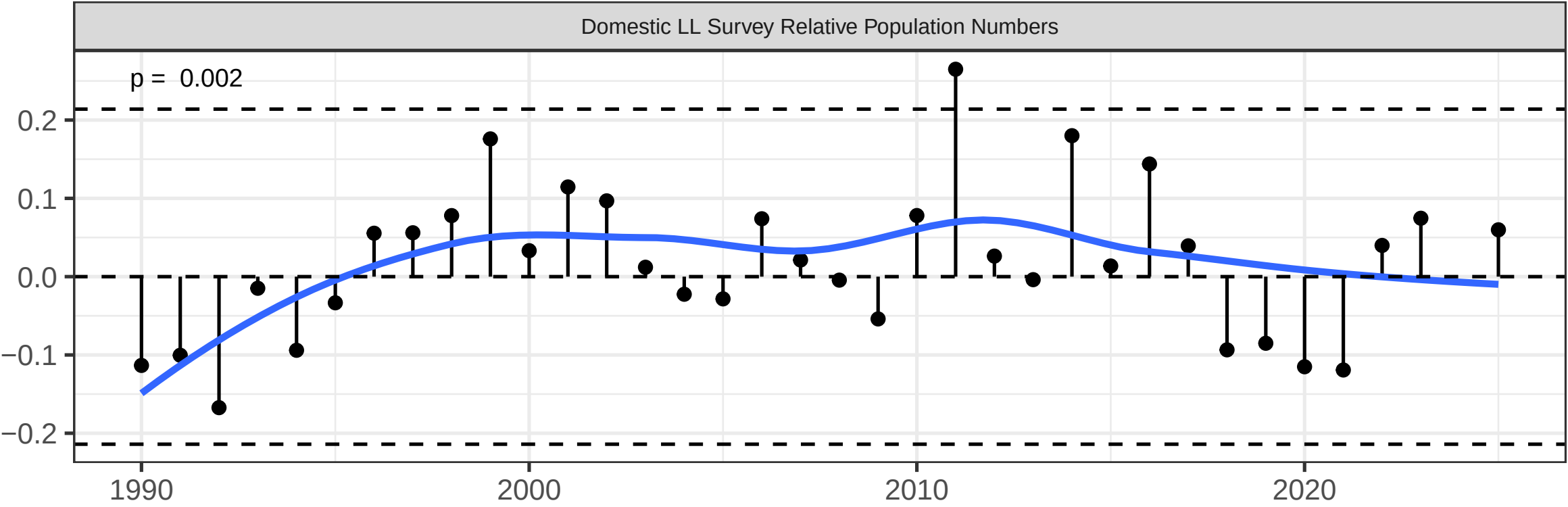
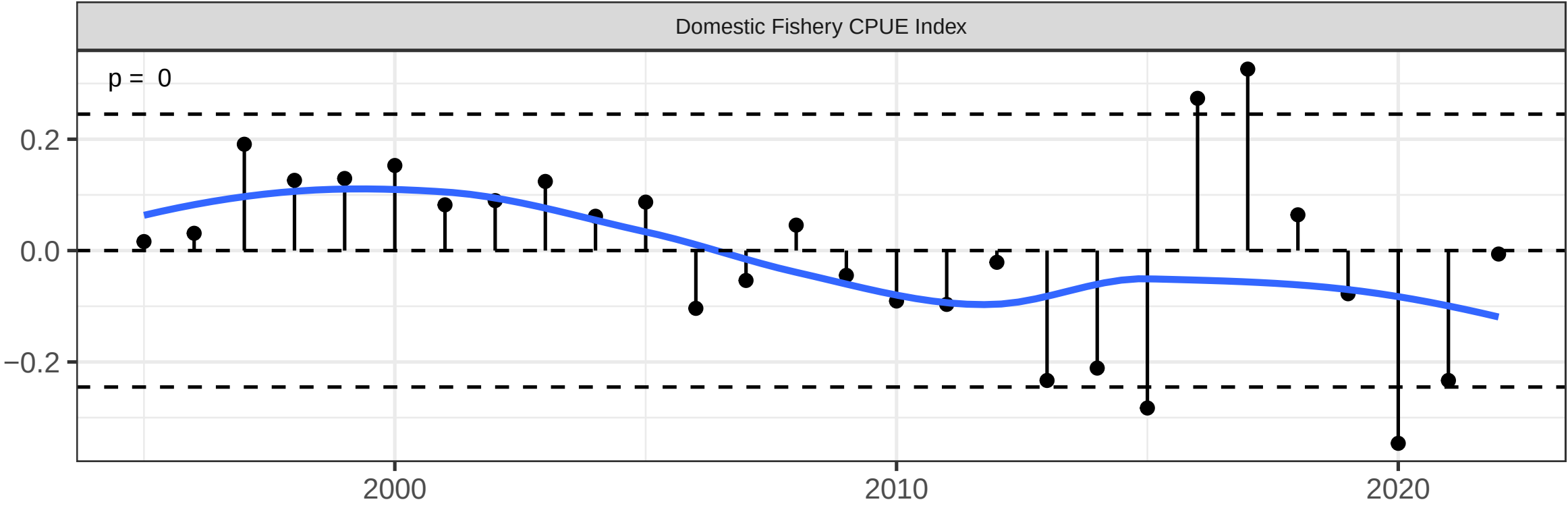




Index Fits

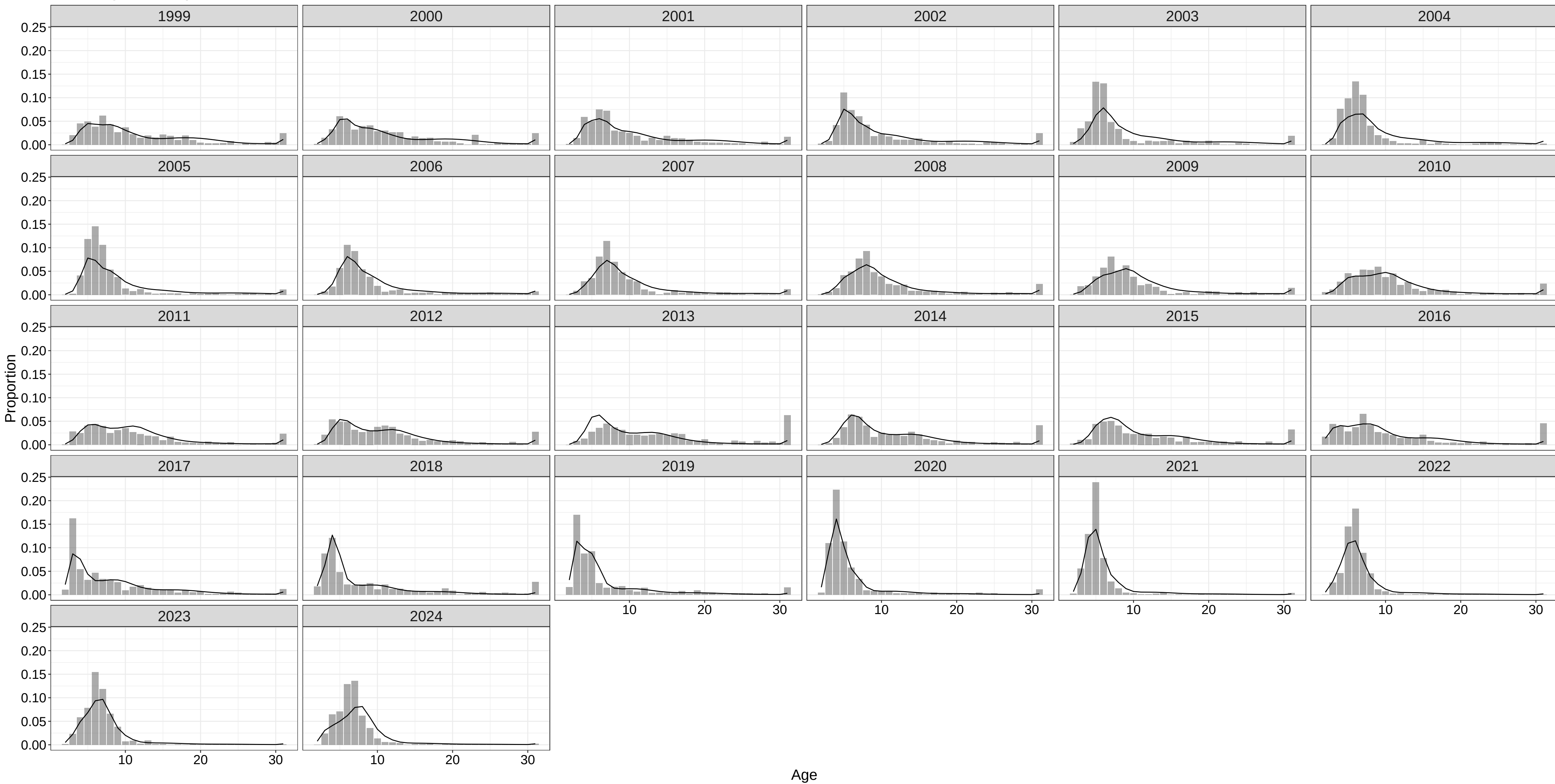


Index Residuals Runs Test

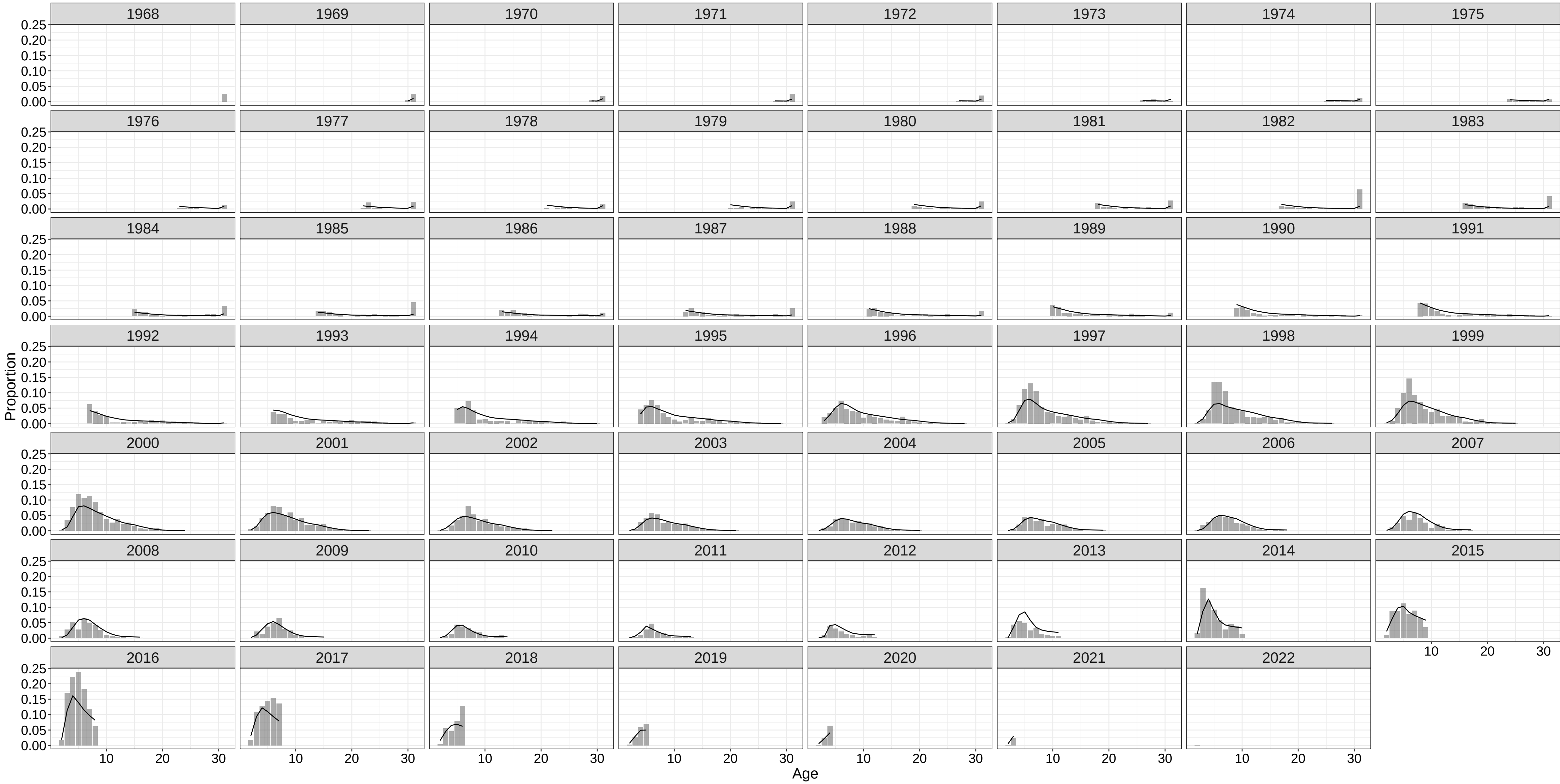


Year

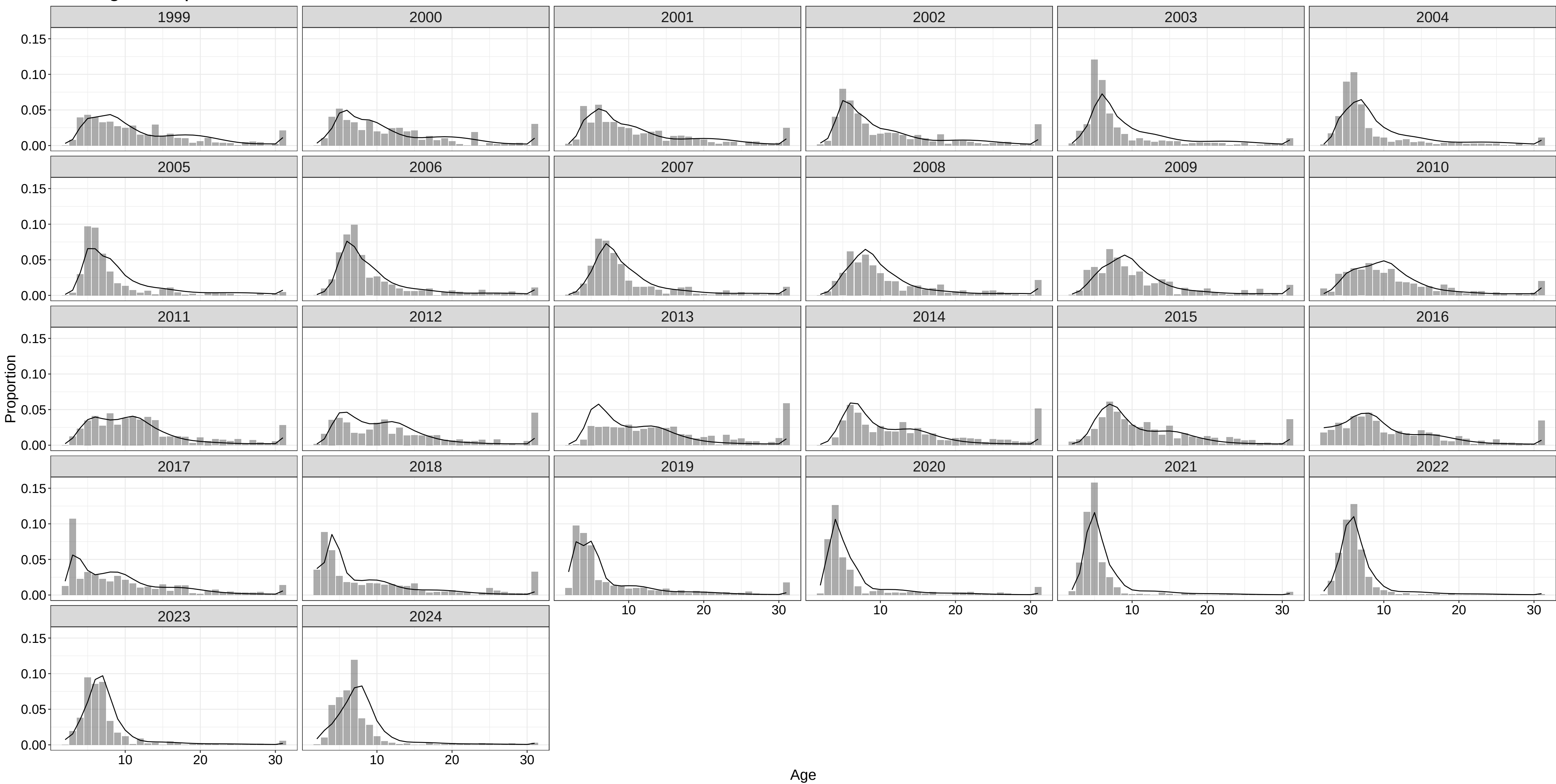
LLF Age Comps Female



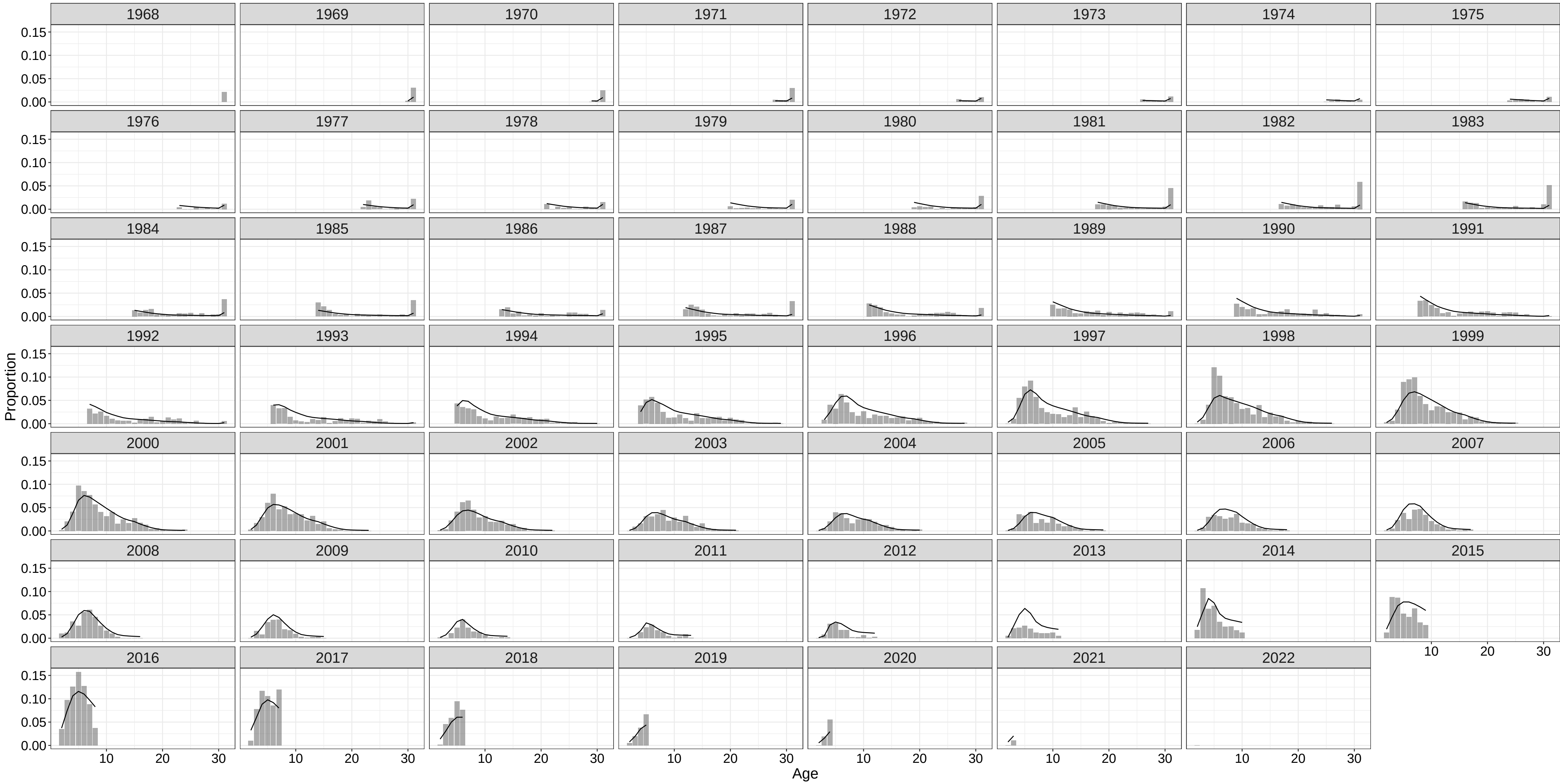
LLF Age Comps Cohort Fits Female



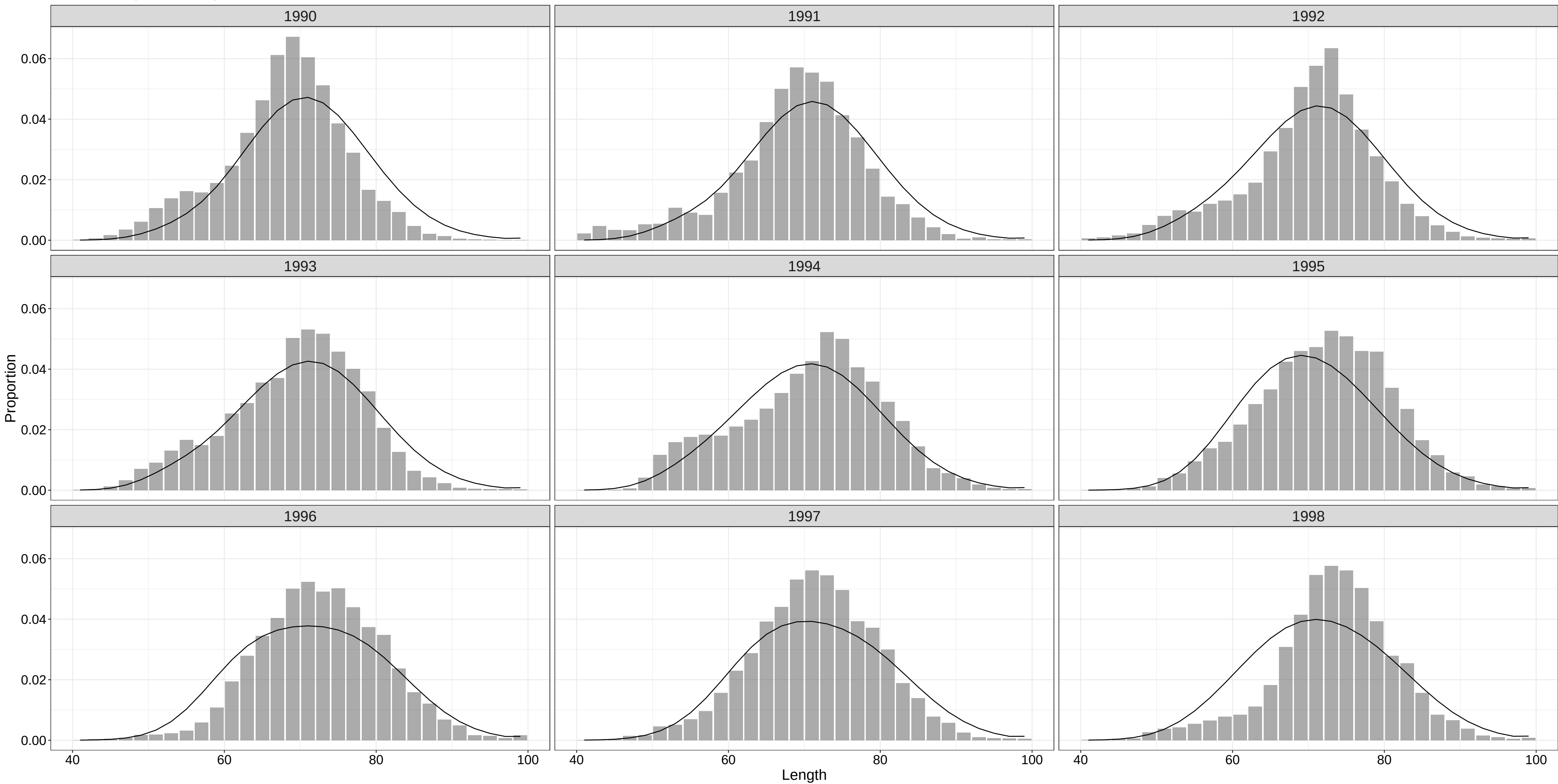
LLF Age Comps Male



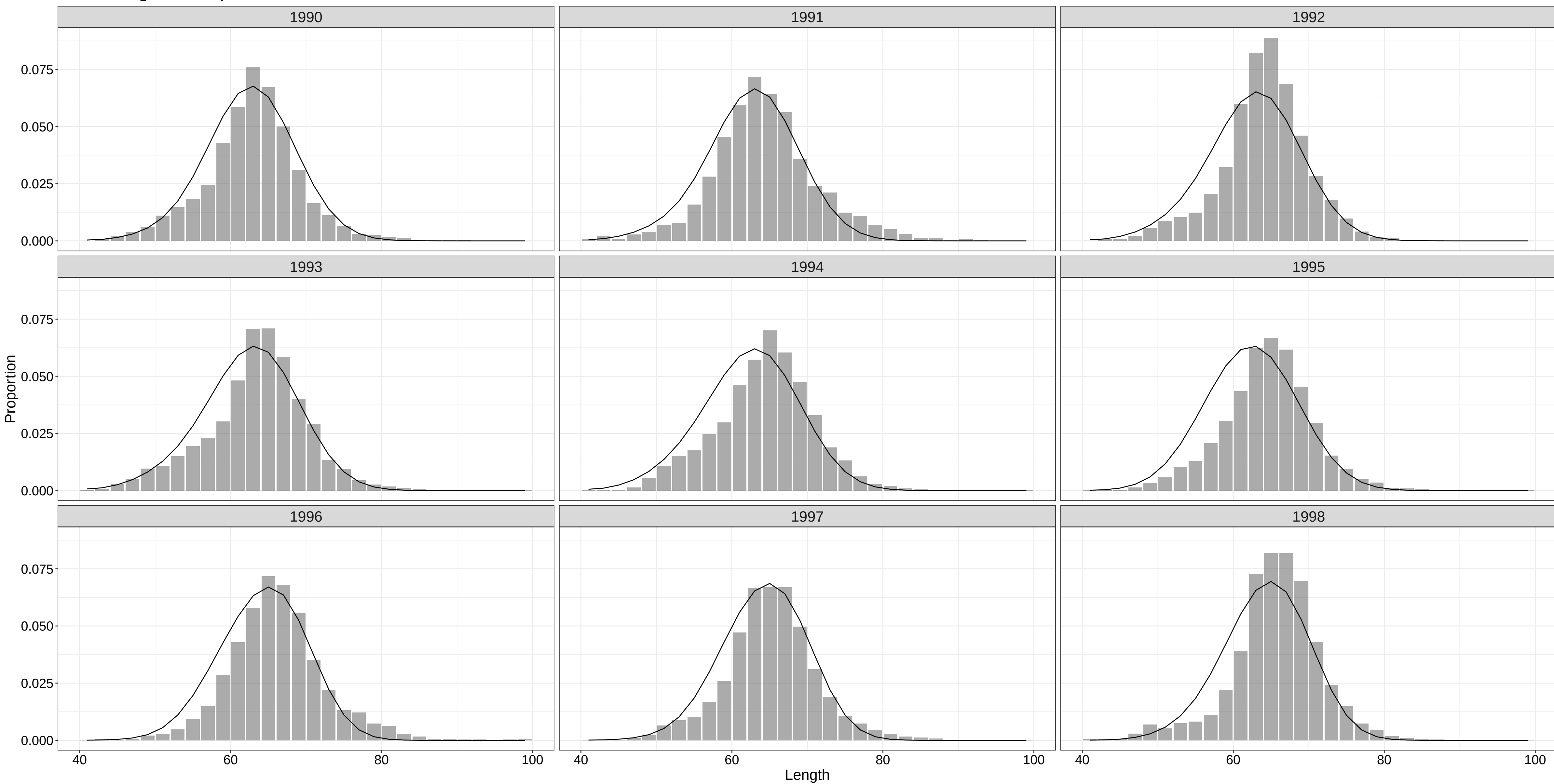
LLF Age Comps Cohort Fits Male



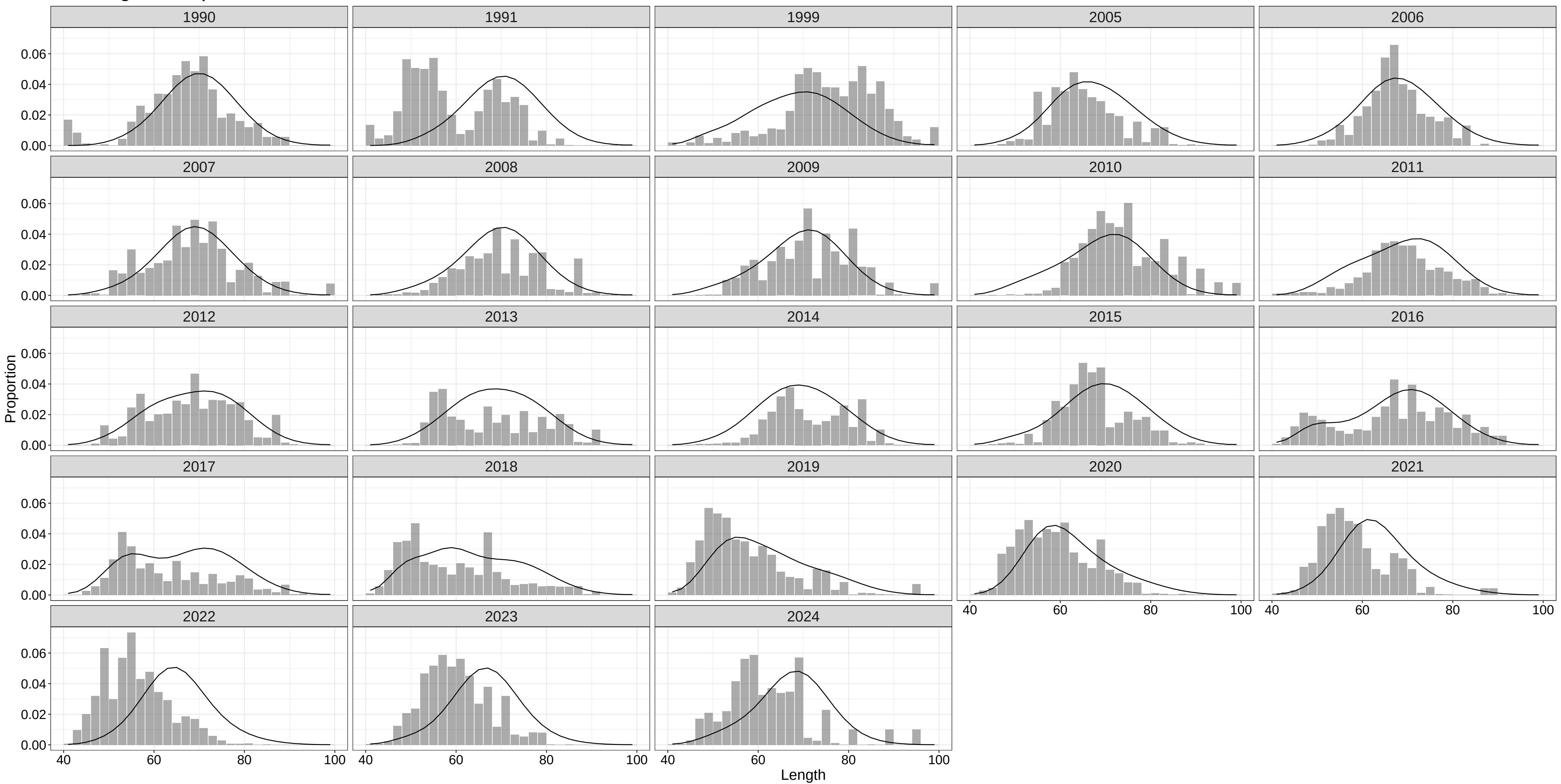
LLF Length Comps Female



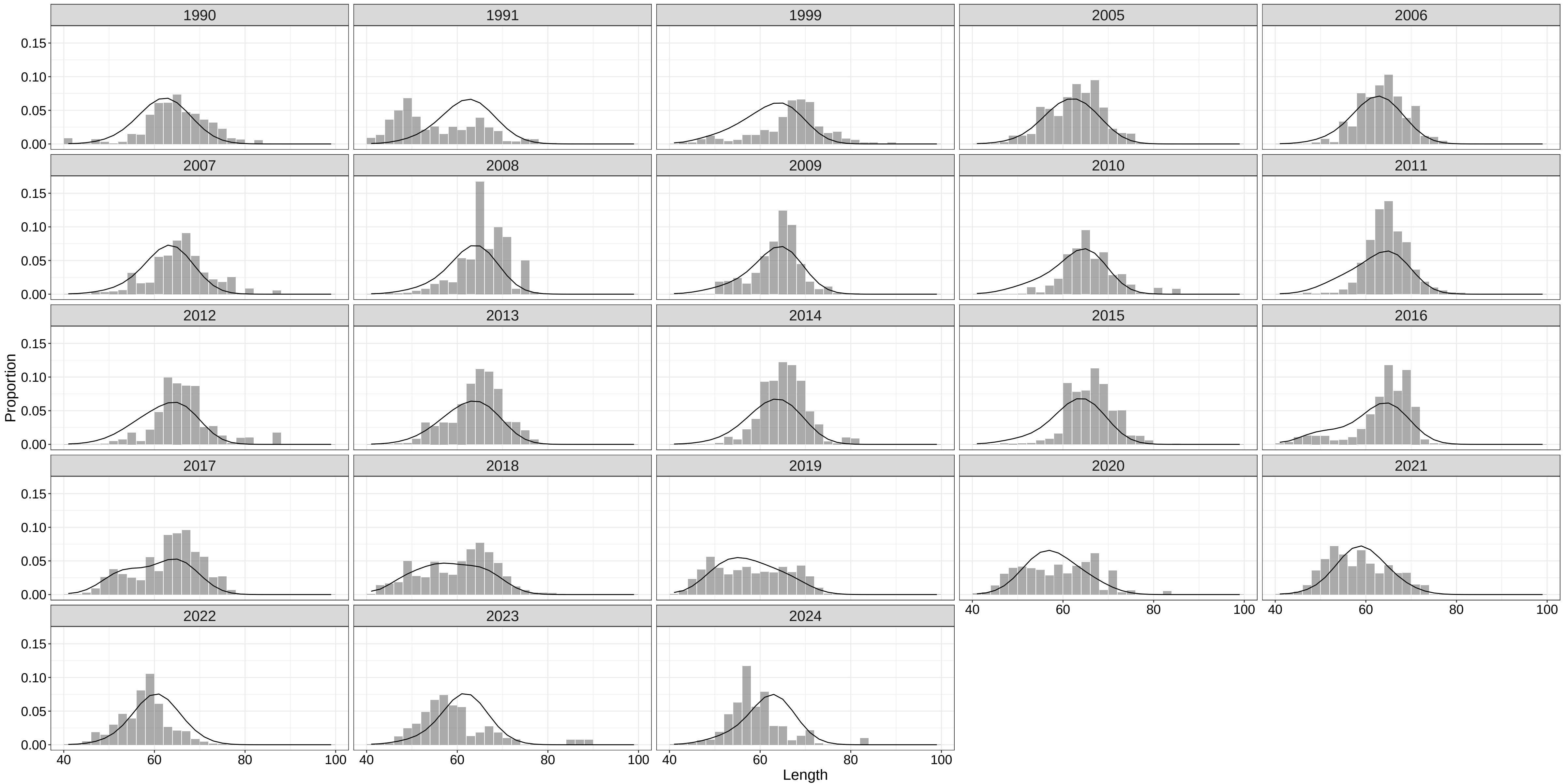
LLF Length Comps Male



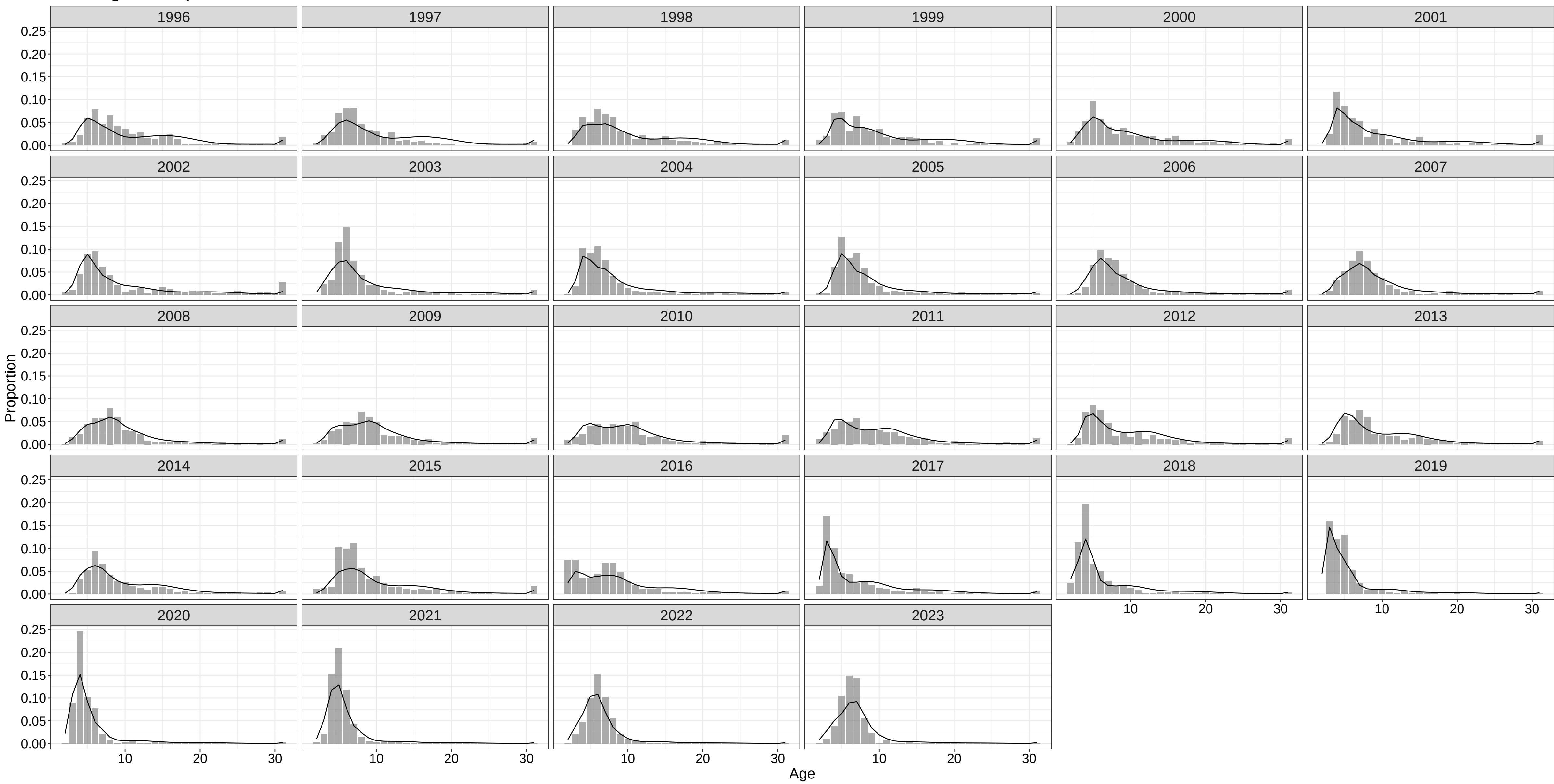
TF Length Comps Female



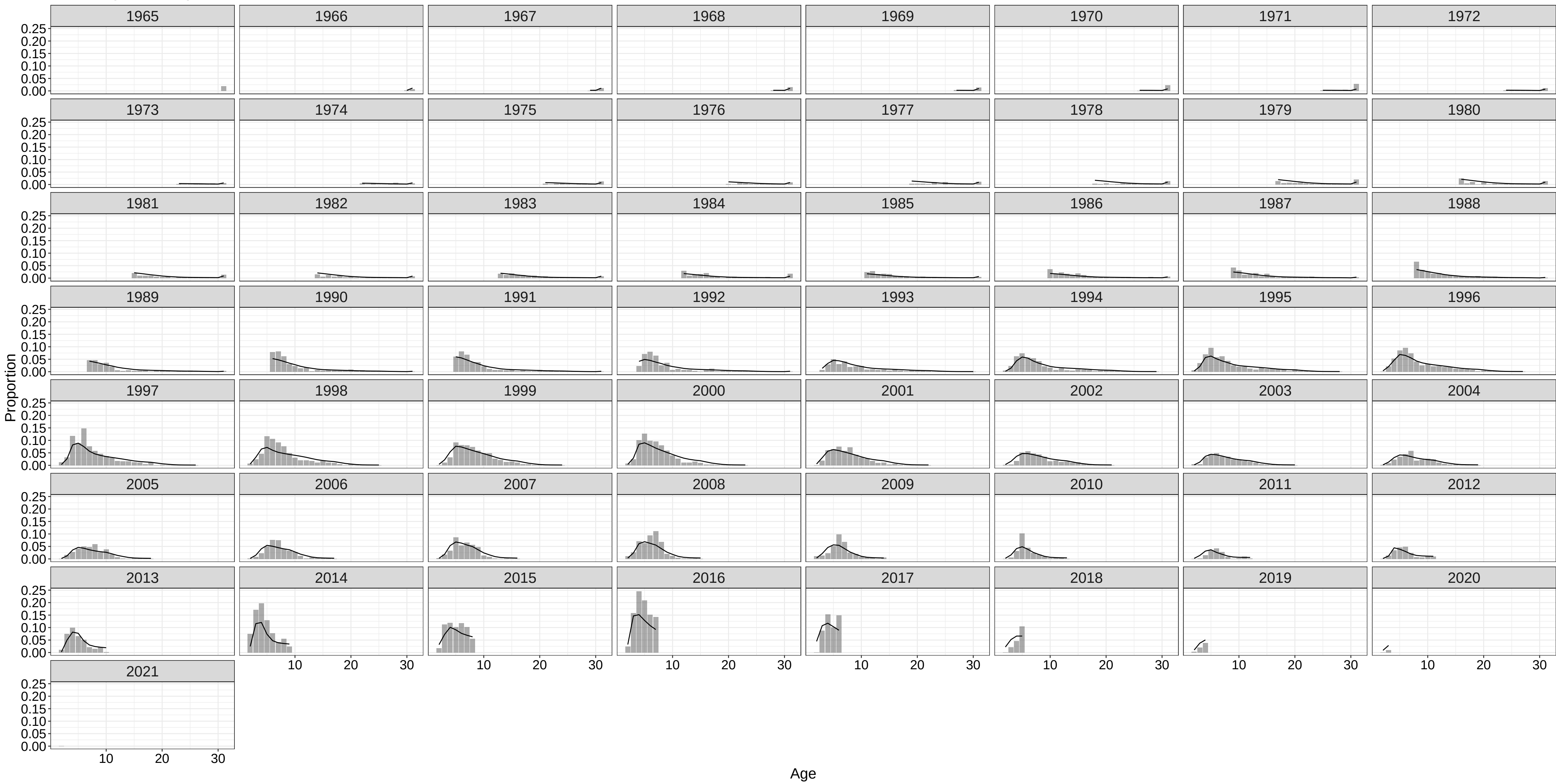
TF Length Comps Male



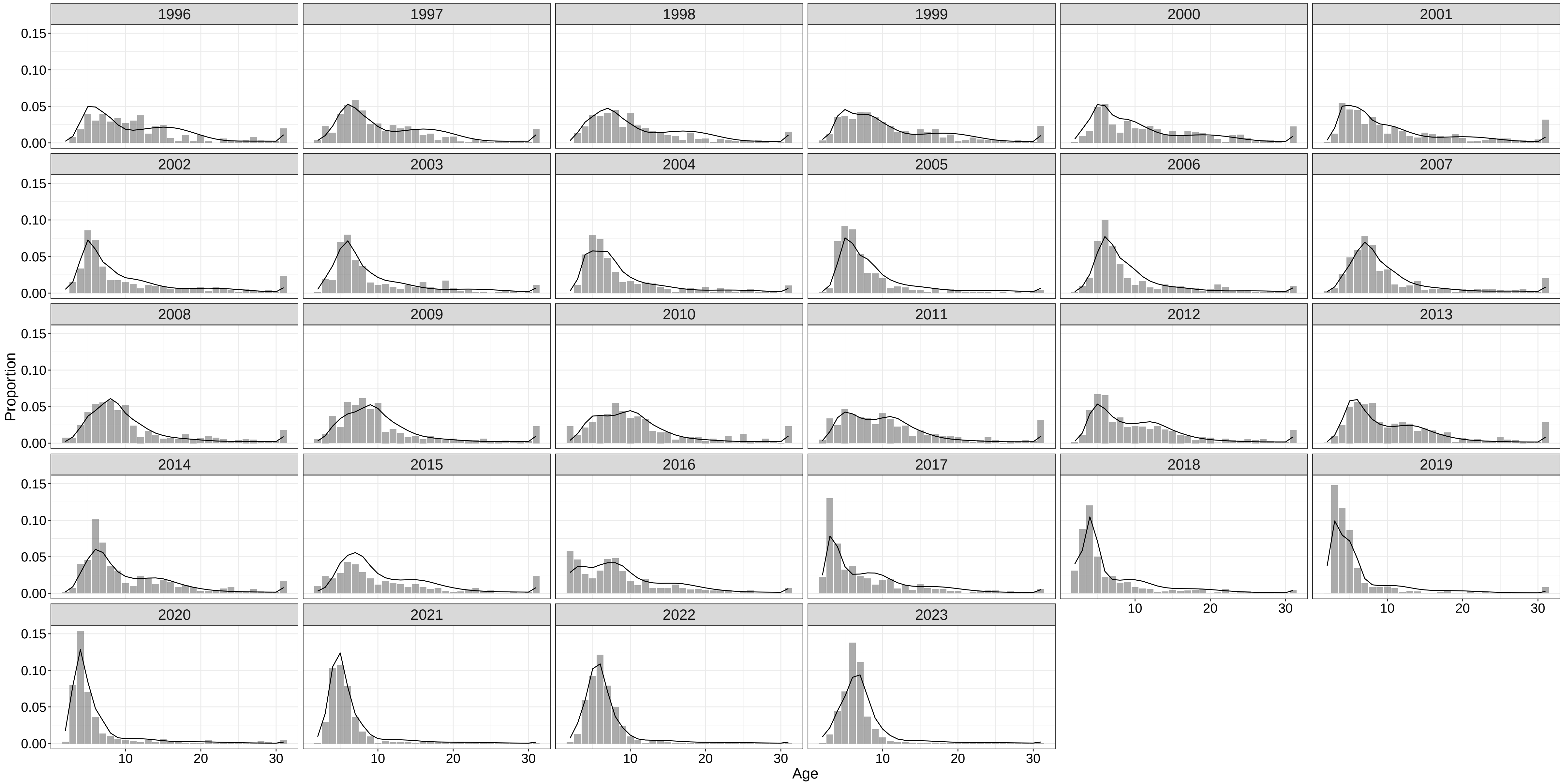
LLS Age Comps Female



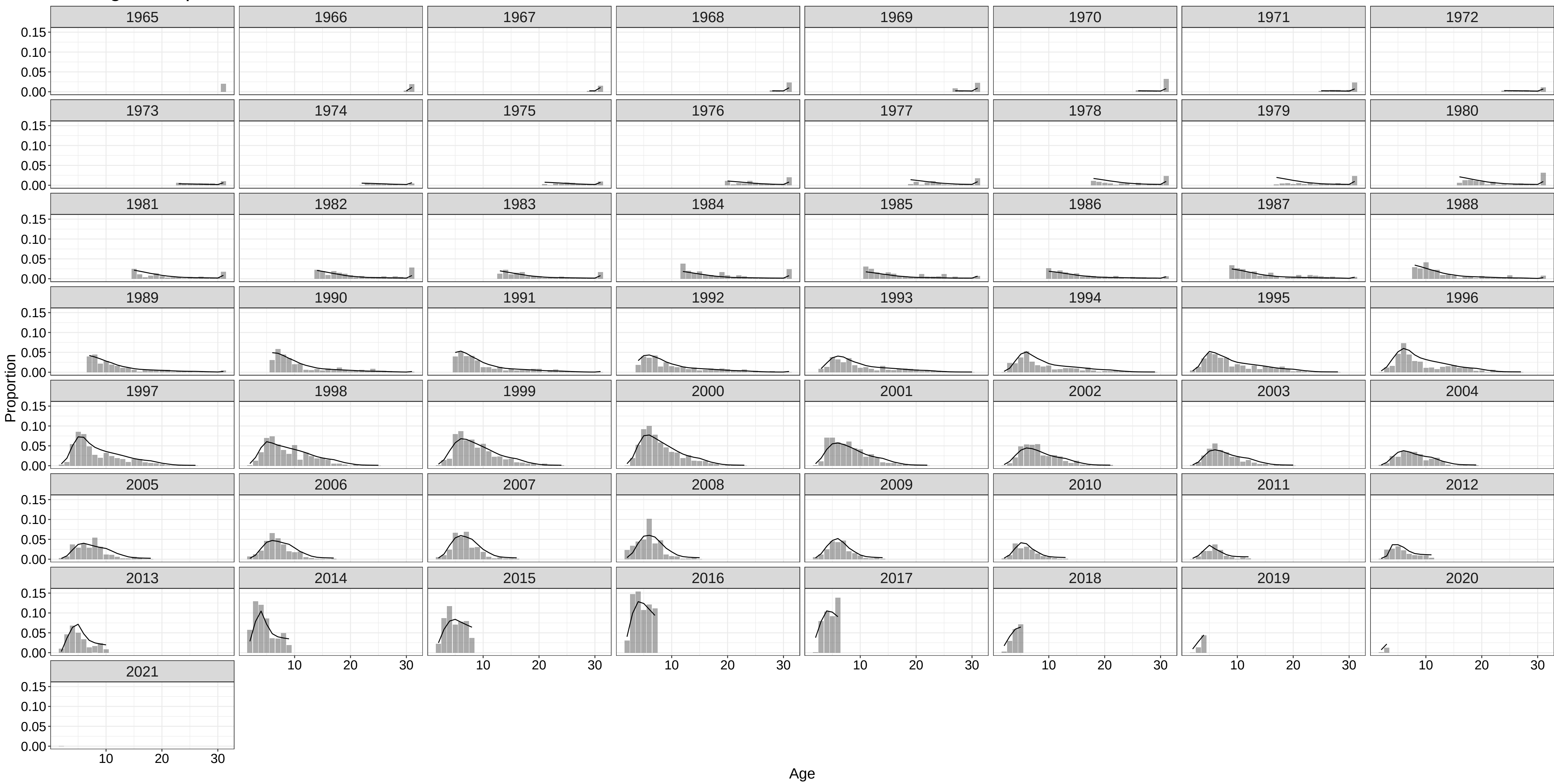
LLS Age Comps Cohort Fit Female



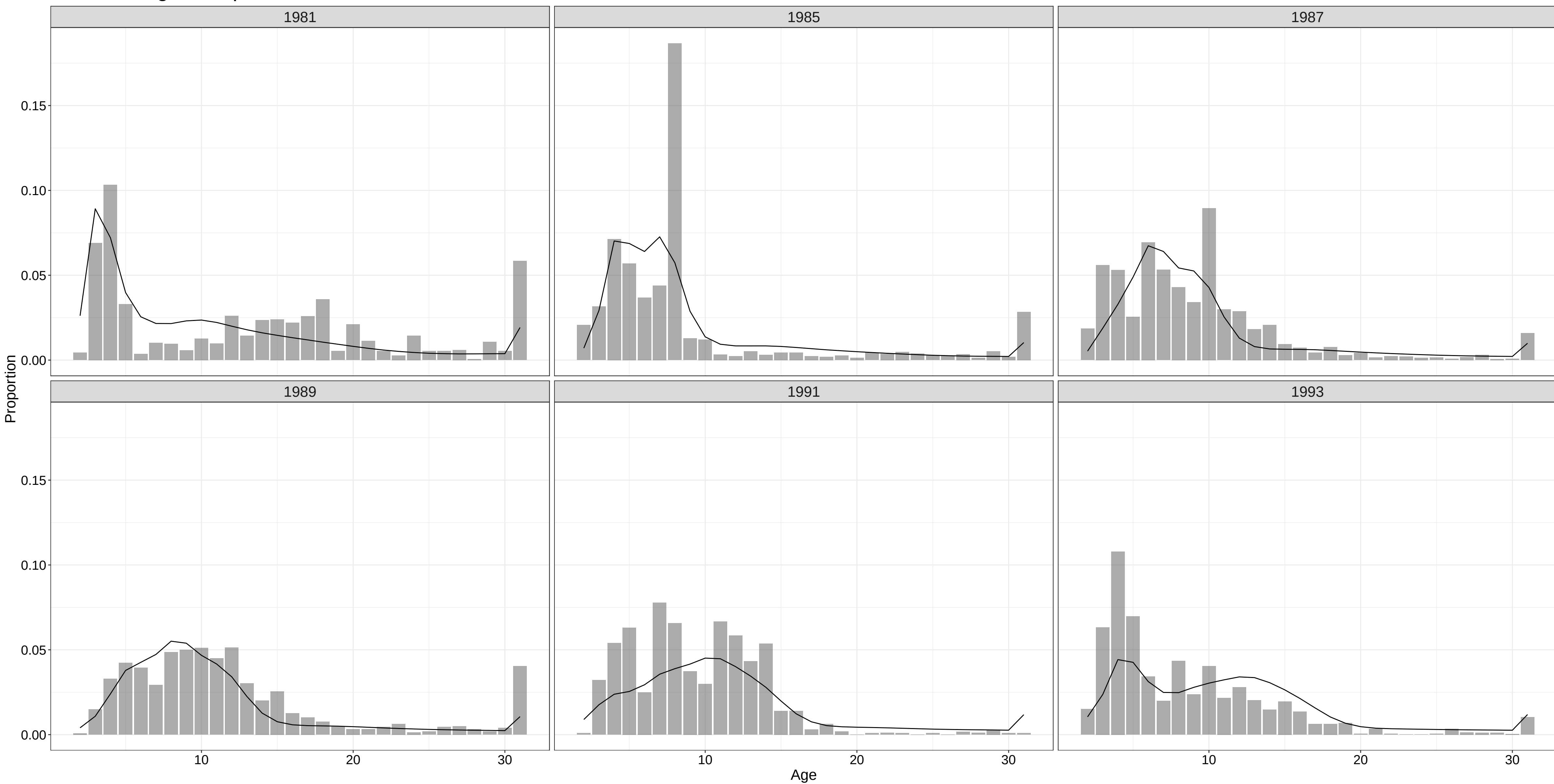
LLS Age Comps Male



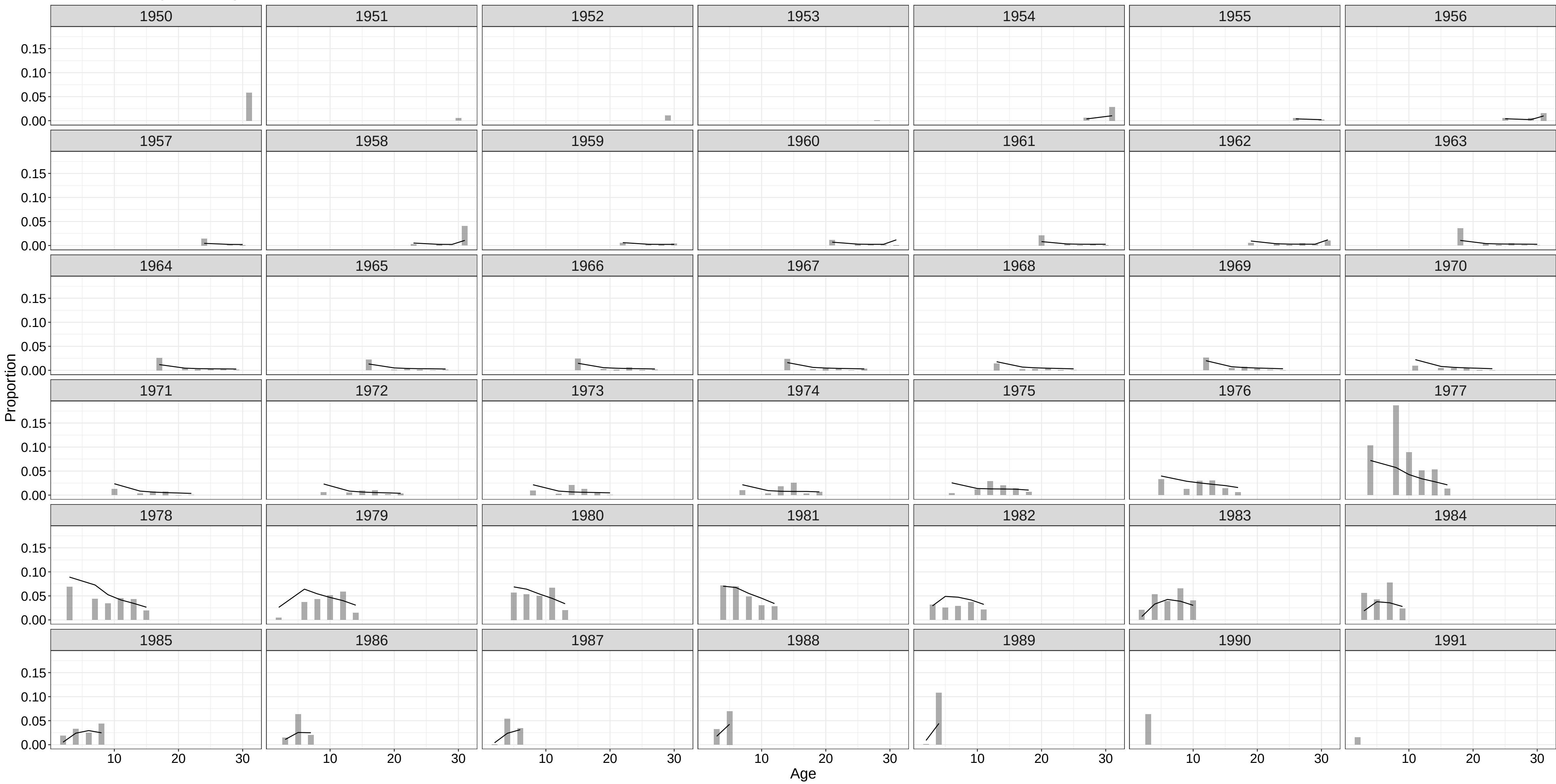
LLS Age Comps Cohort Fit Male



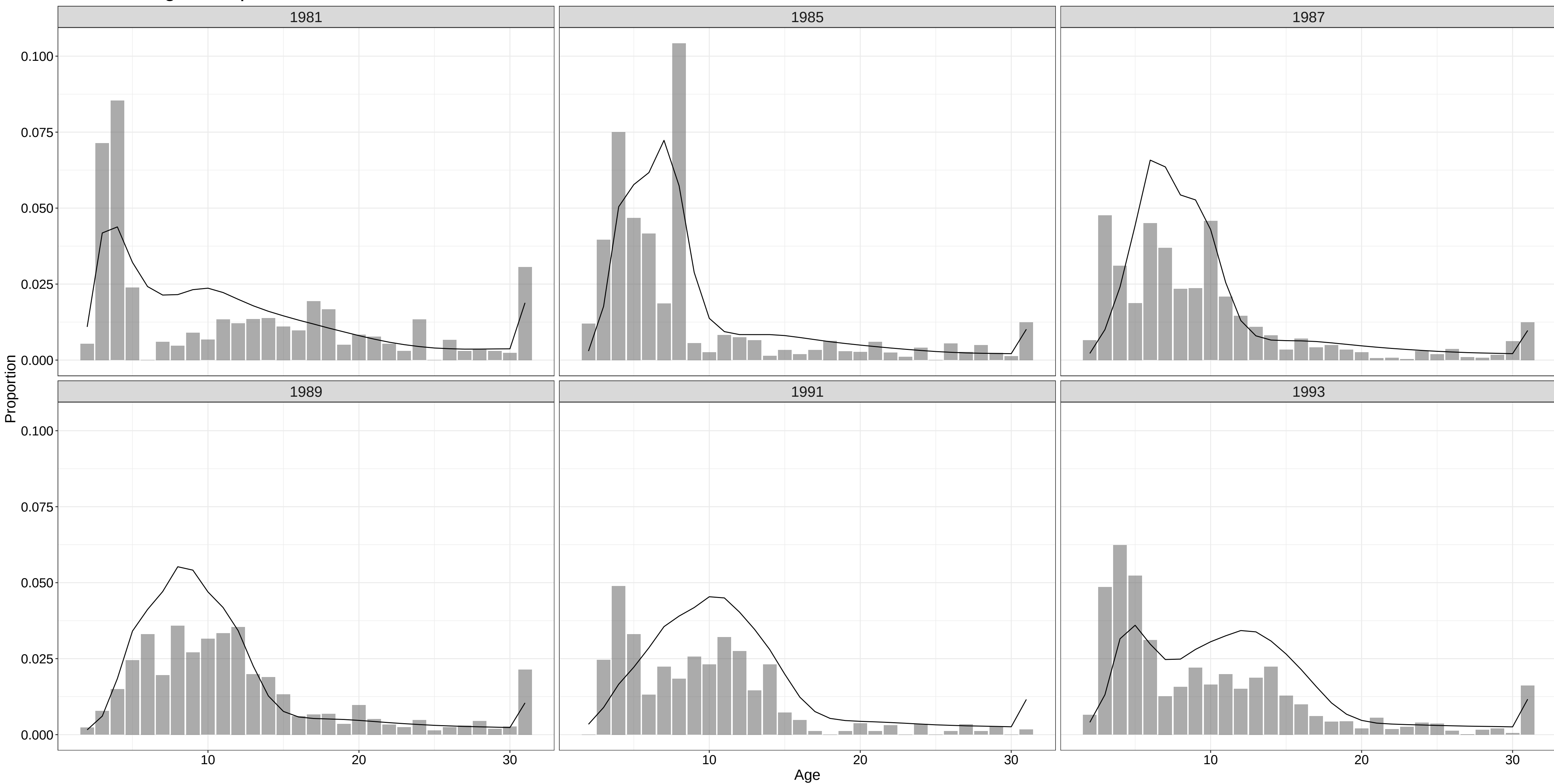
JPN LLS Age Comps Female



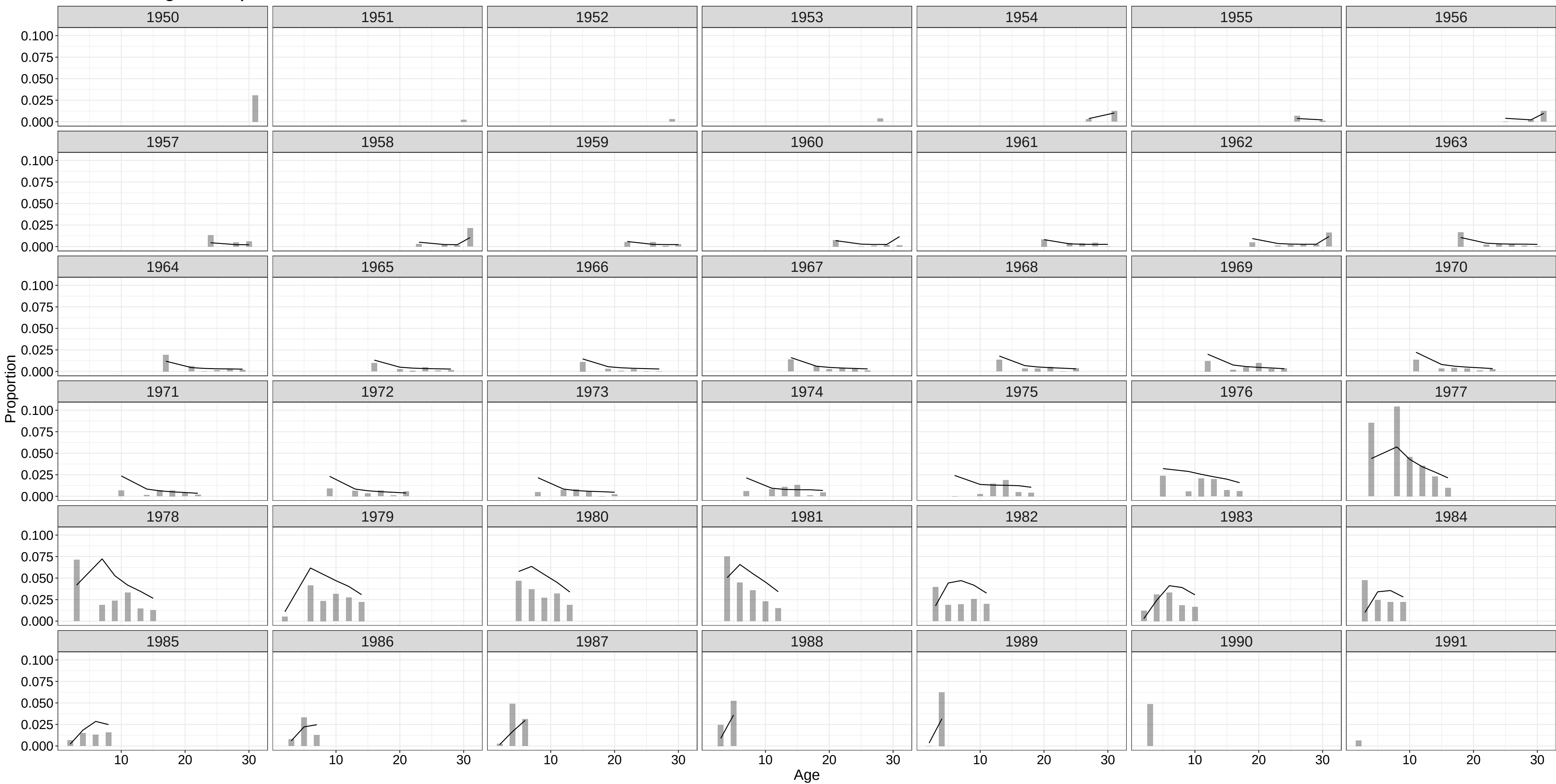
JPN LLS Age Comps Cohort Fit Female



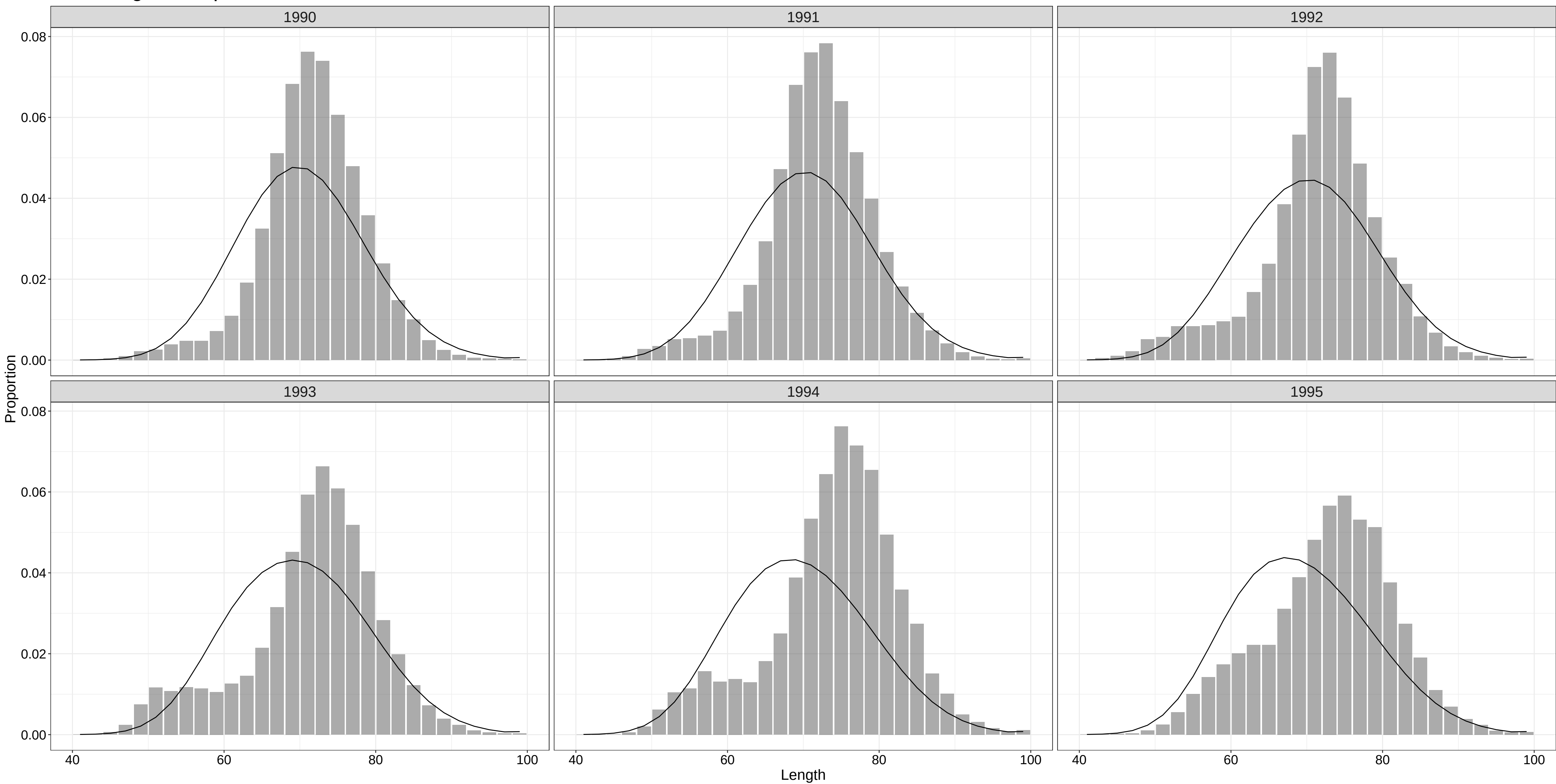
JPN LLS Age Comps Male



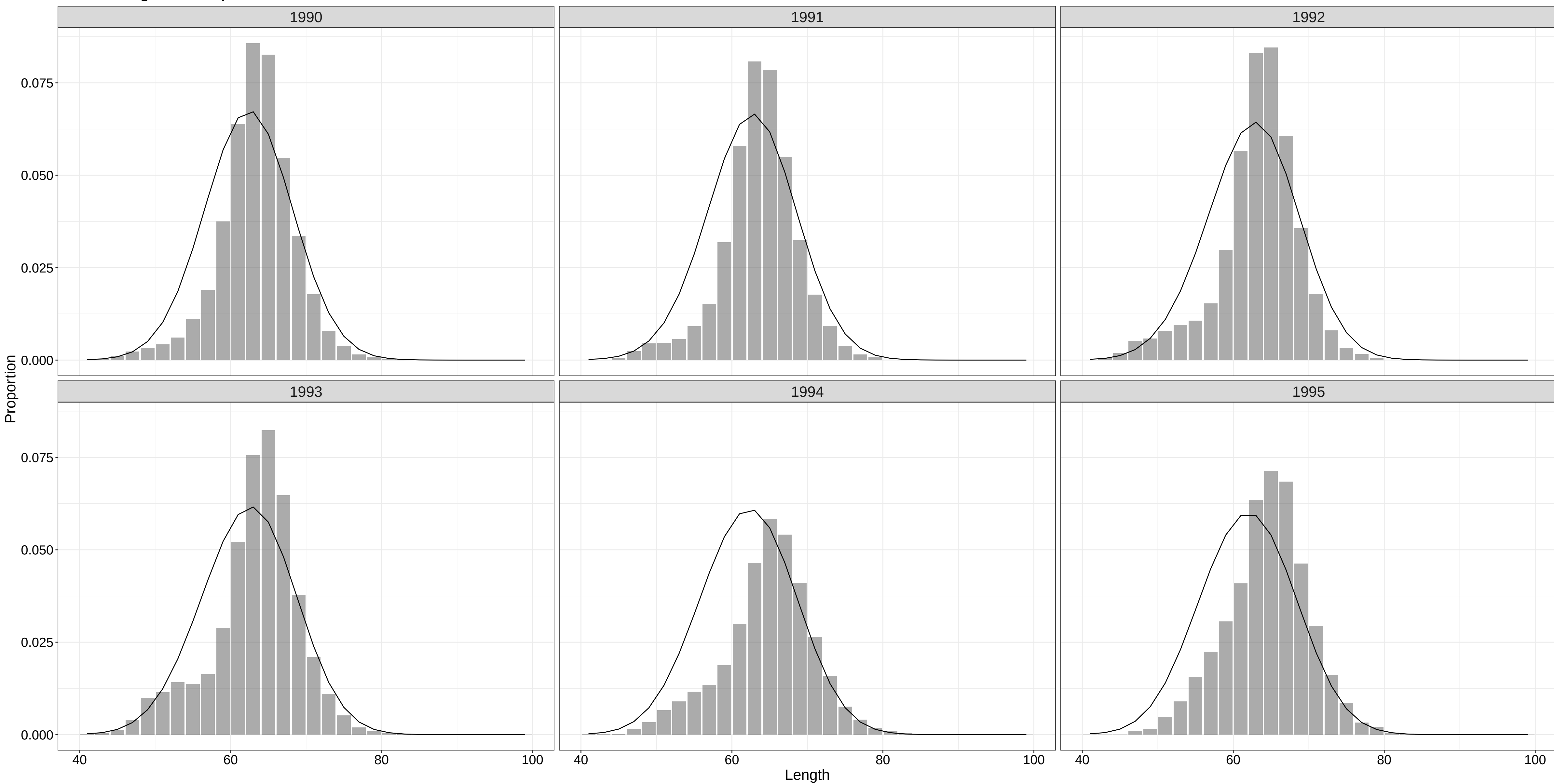
JPN LLS Age Comps Cohort Fit Male



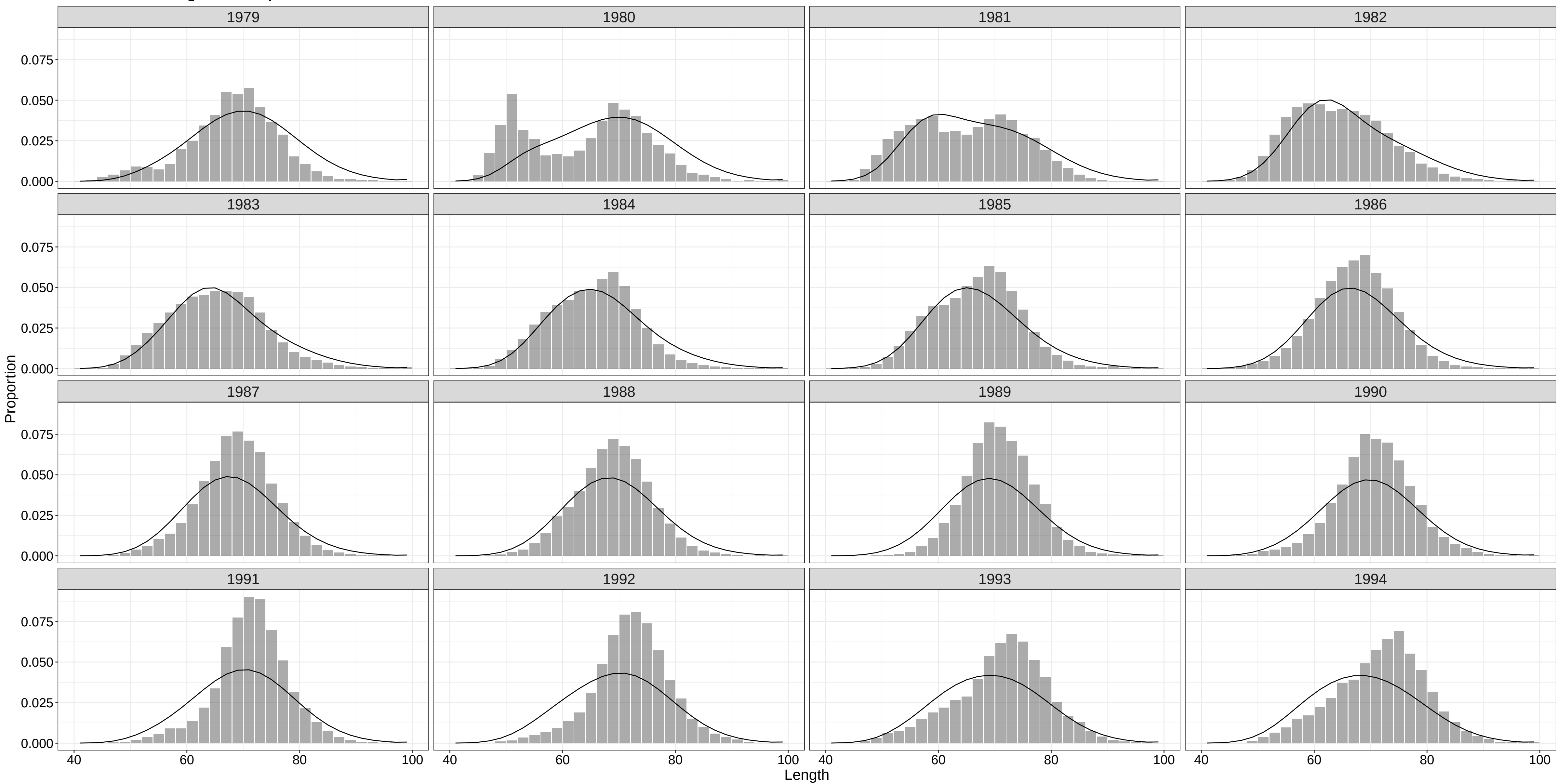
LLS Length Comps Female



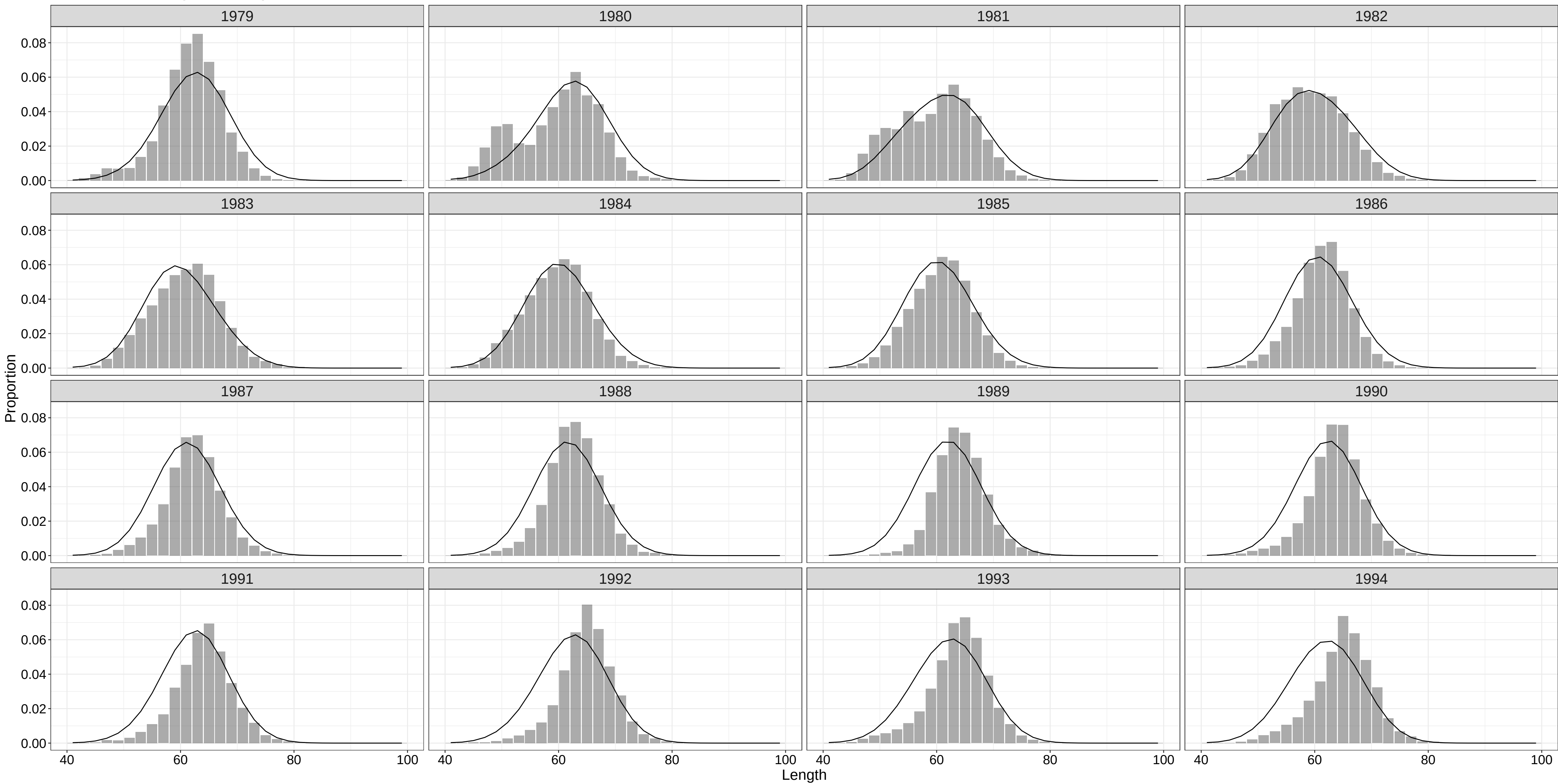
LLS Length Comps Male



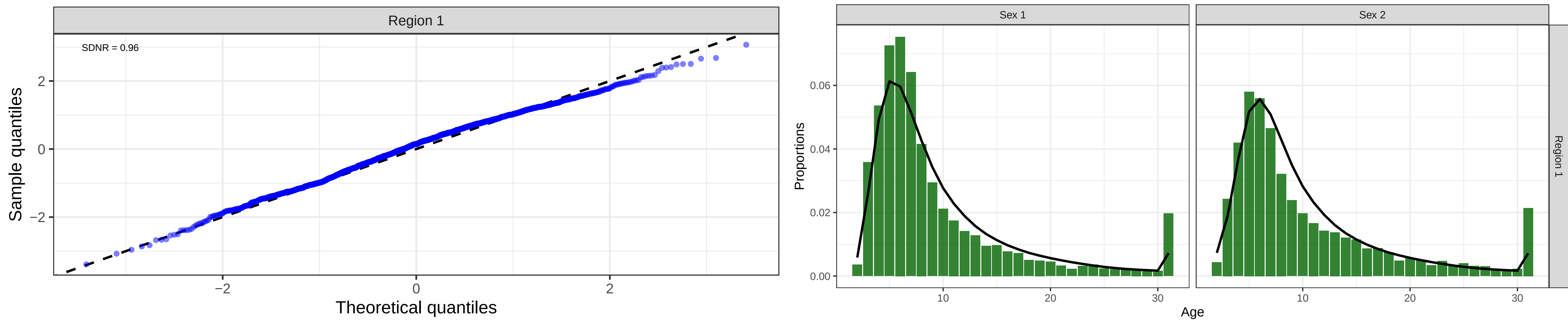
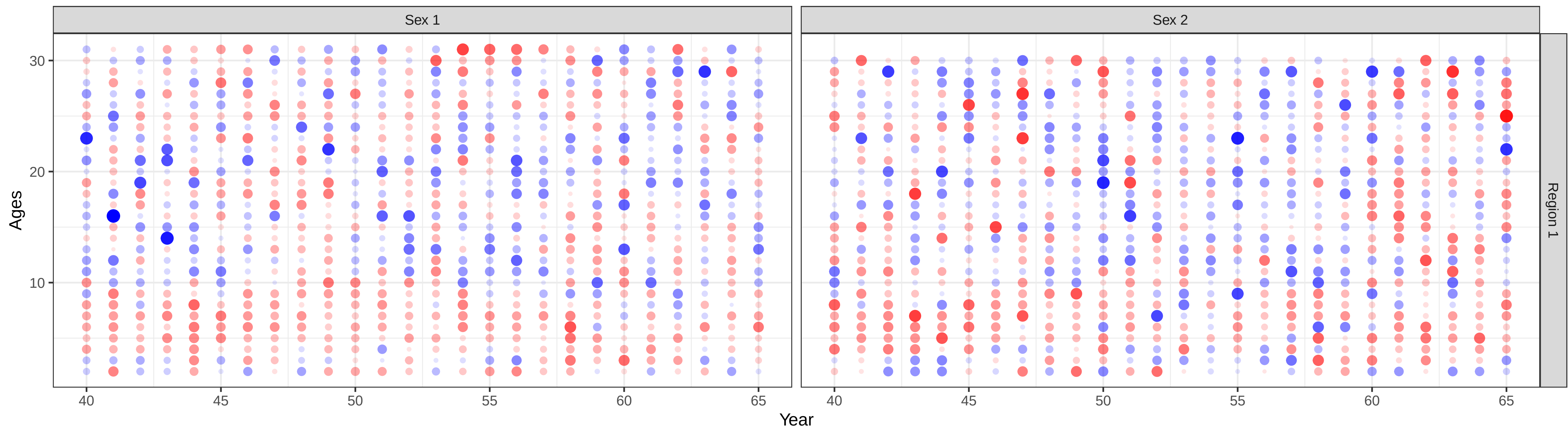
JPN LLS Length Comps Female



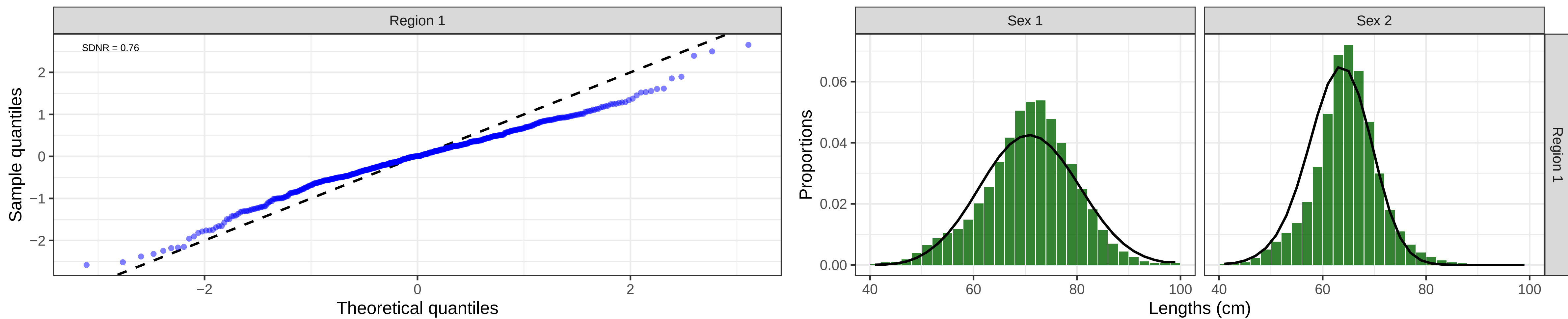
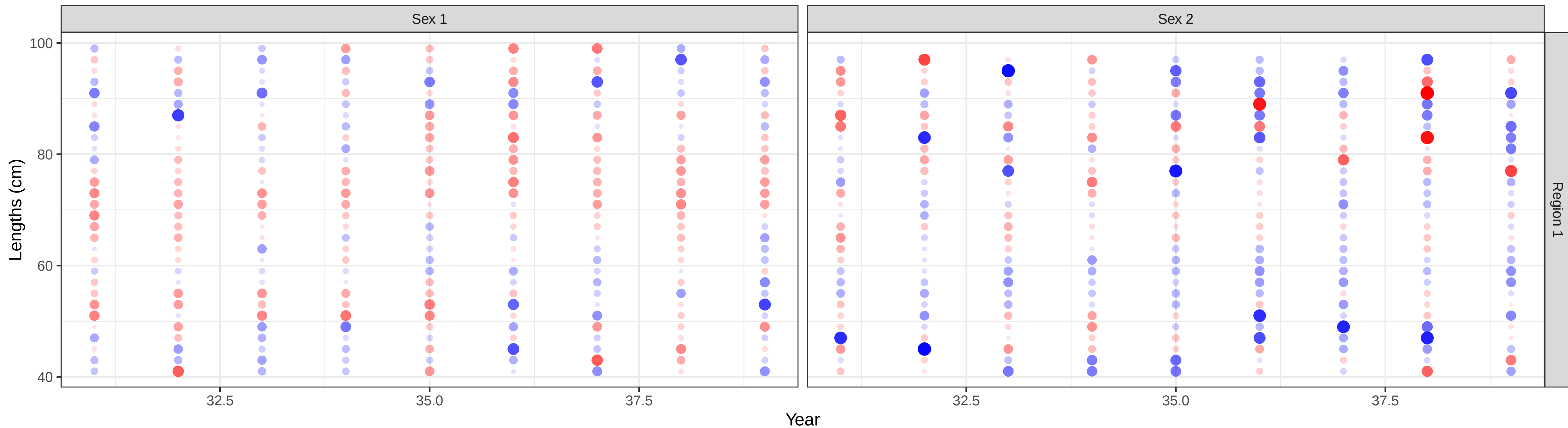
JPN LLS Length Comps Male

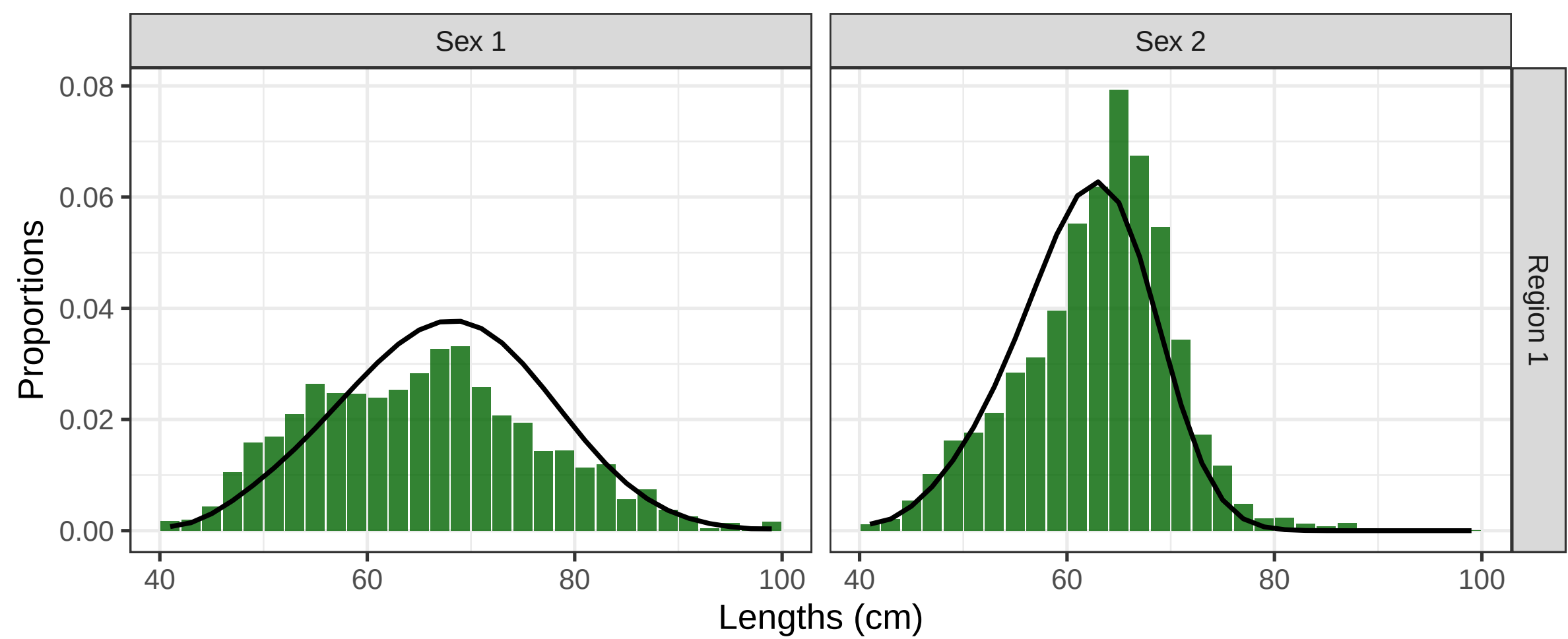
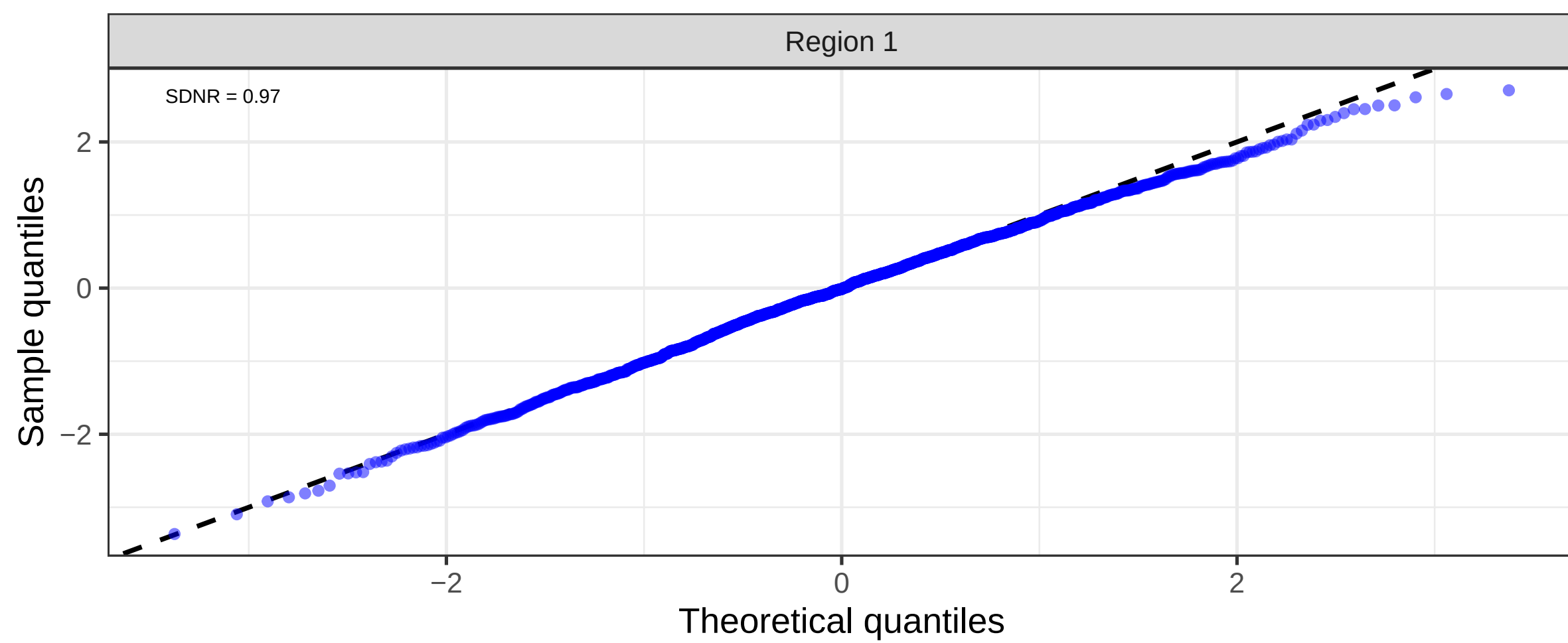
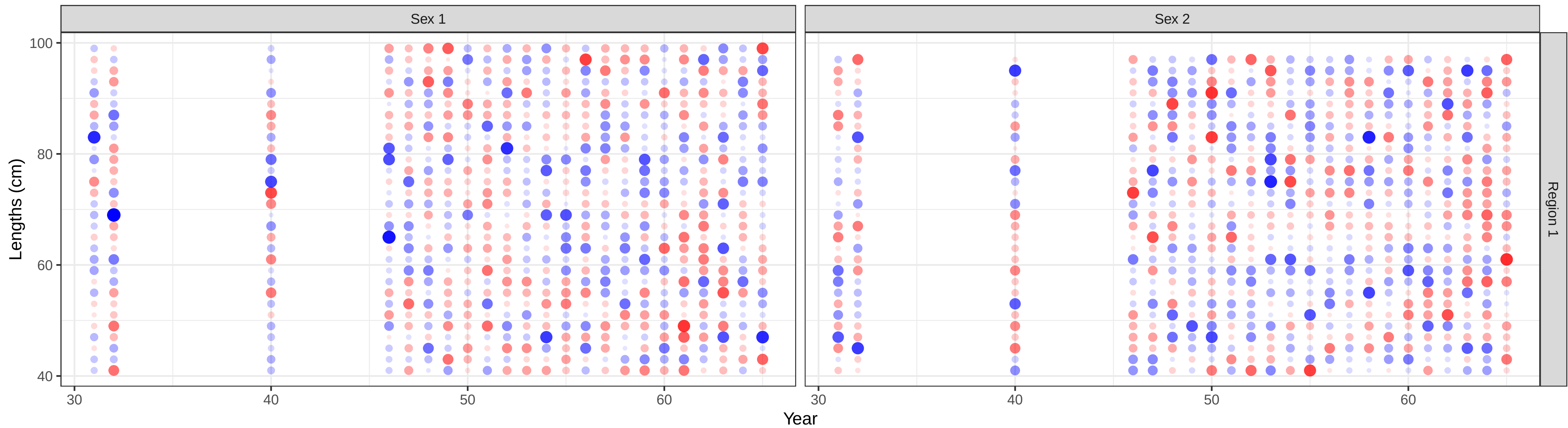


abs(Resid) 1 2 3 Sign ● Neg ● Pos

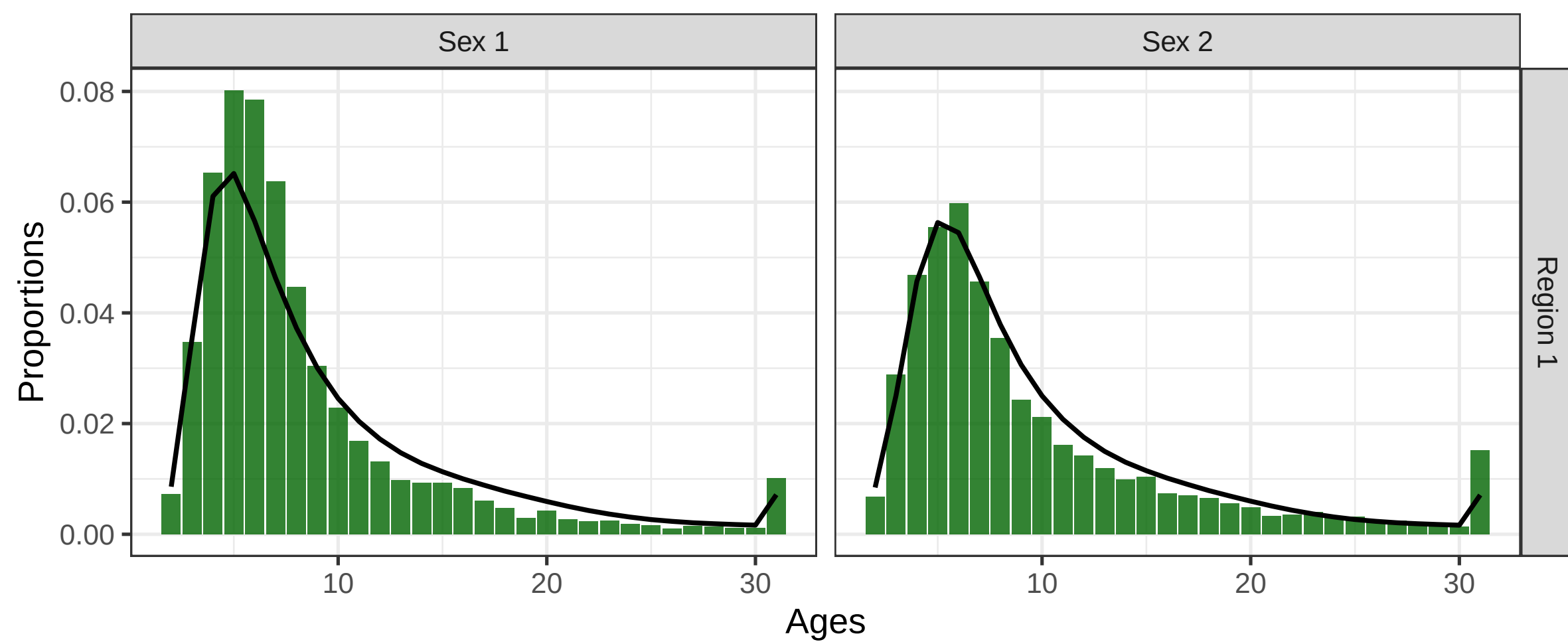
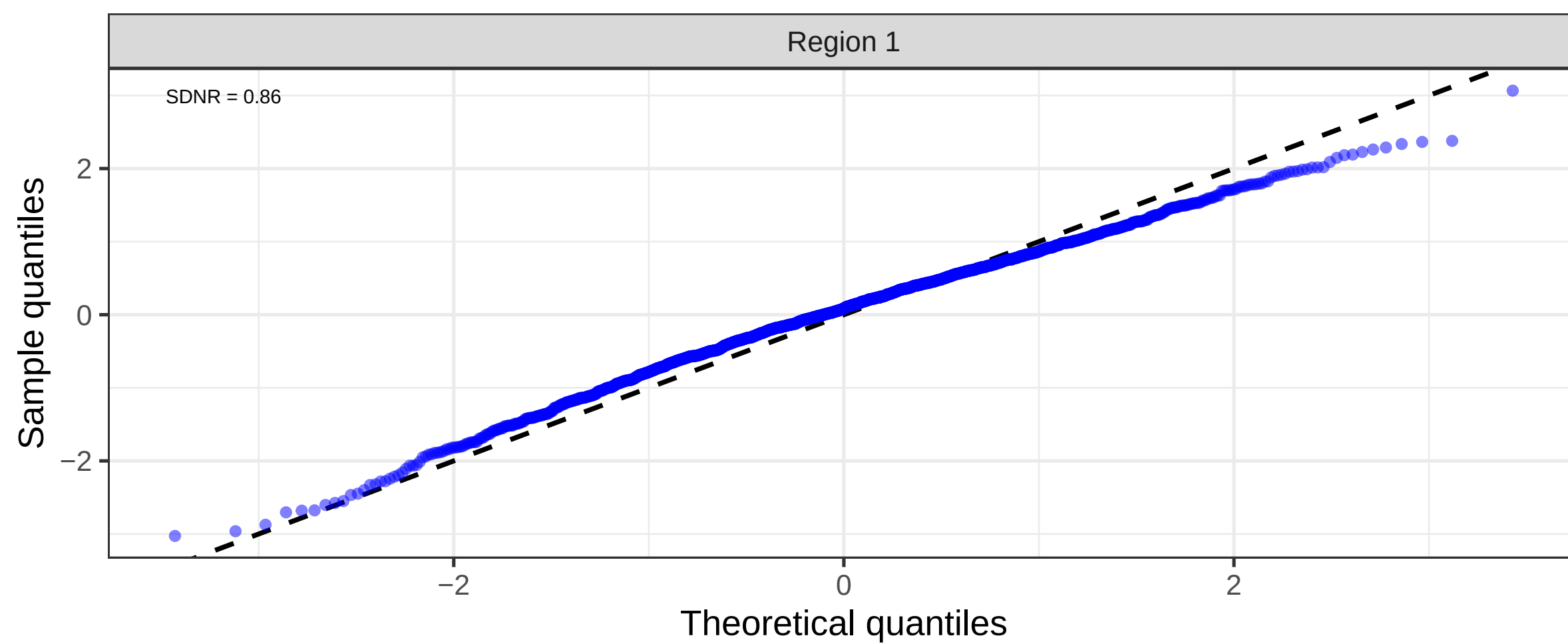
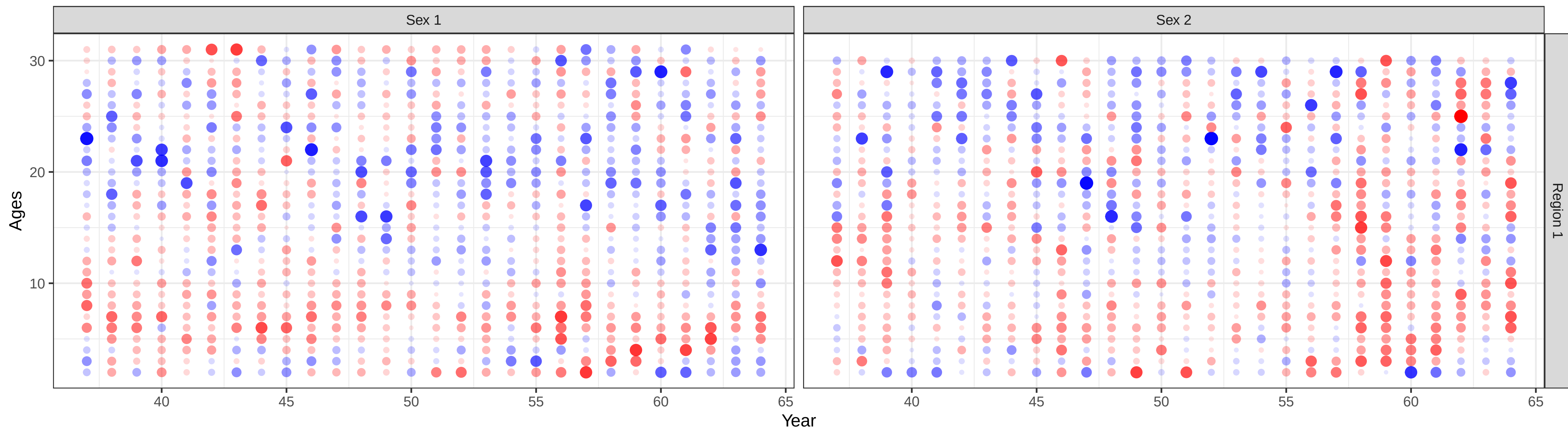


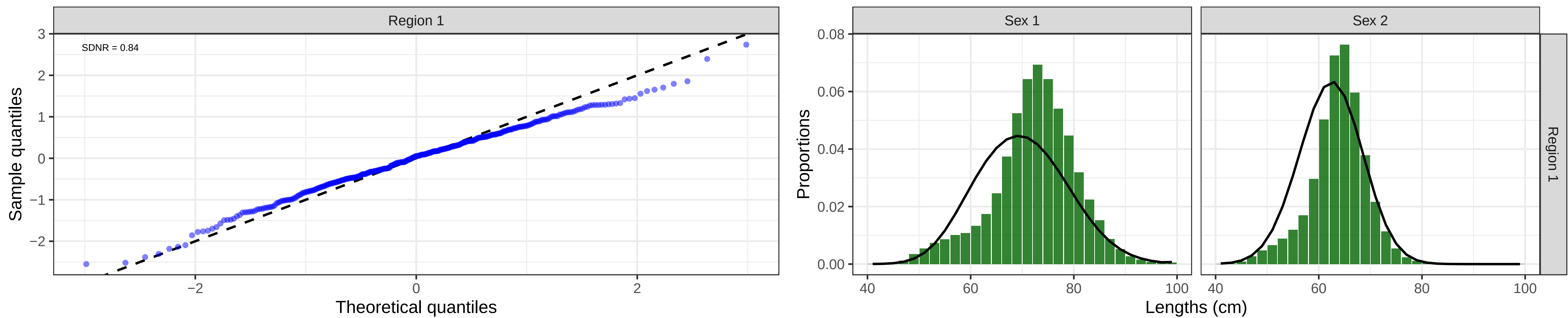
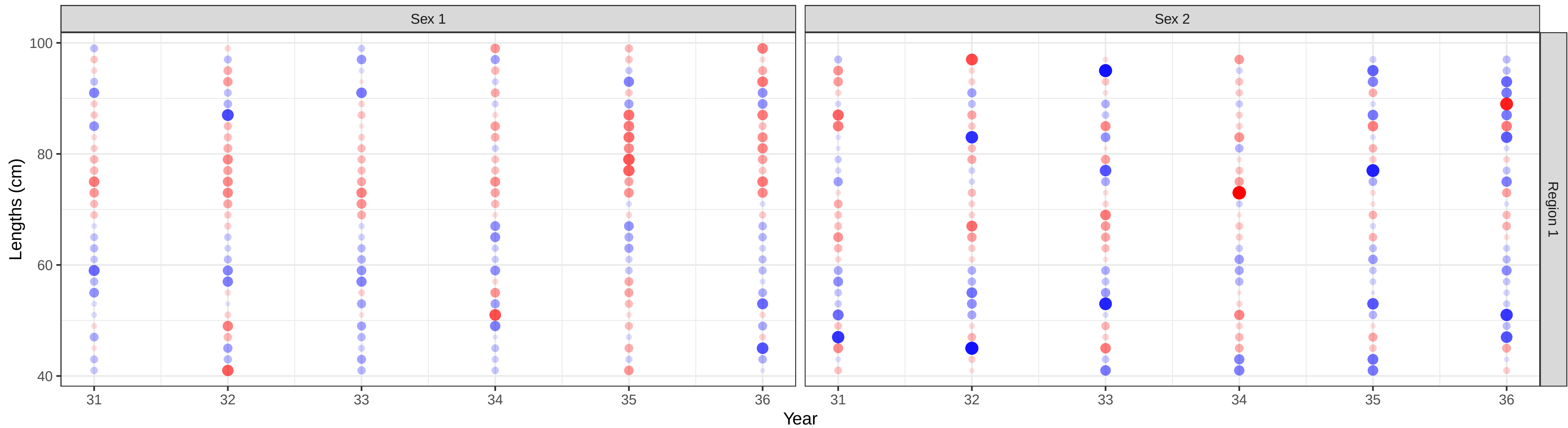
Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0 ● 2.5



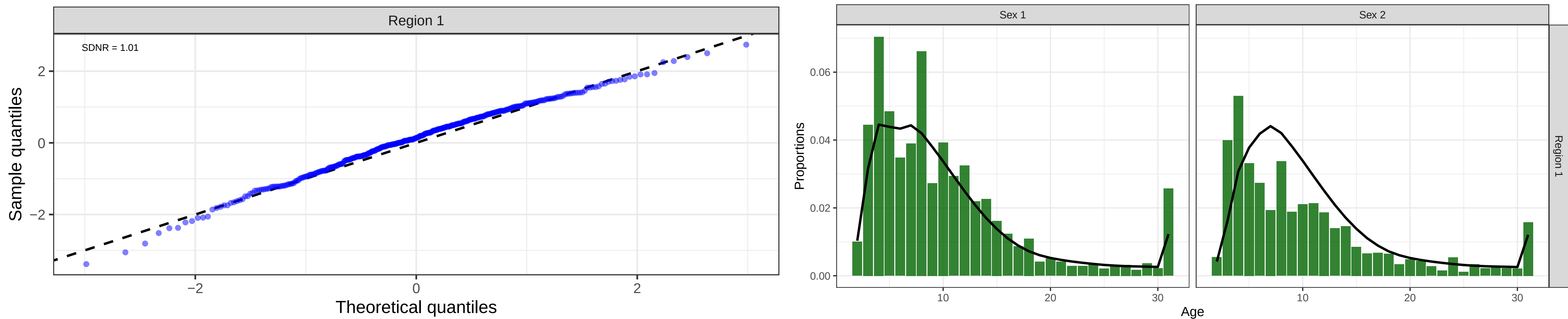
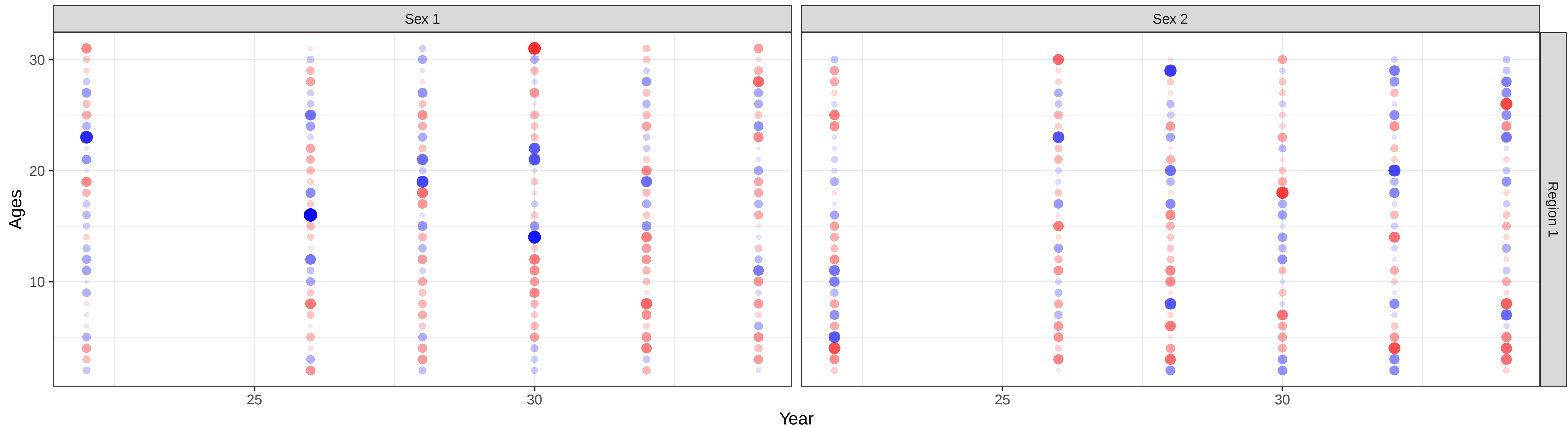


abs(Resid) 1 2 3 Sign ● Neg ● Pos

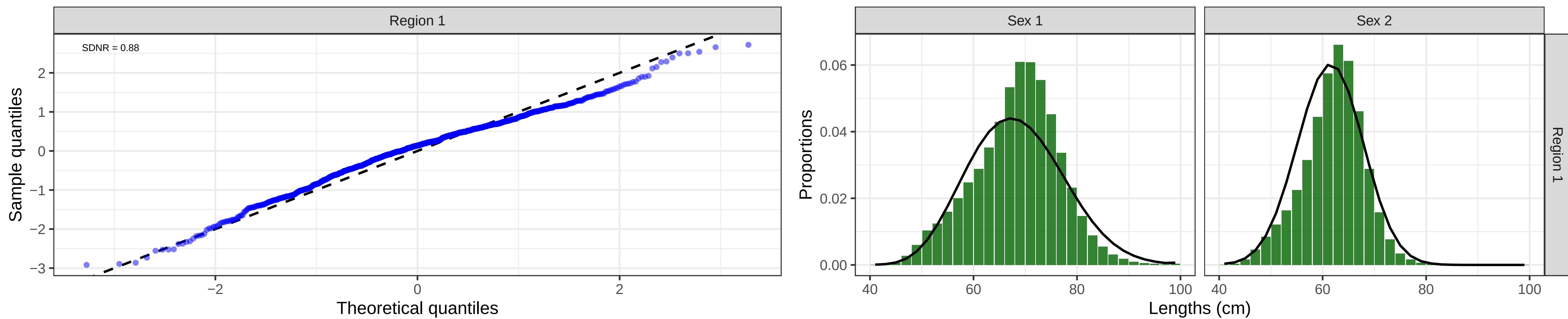
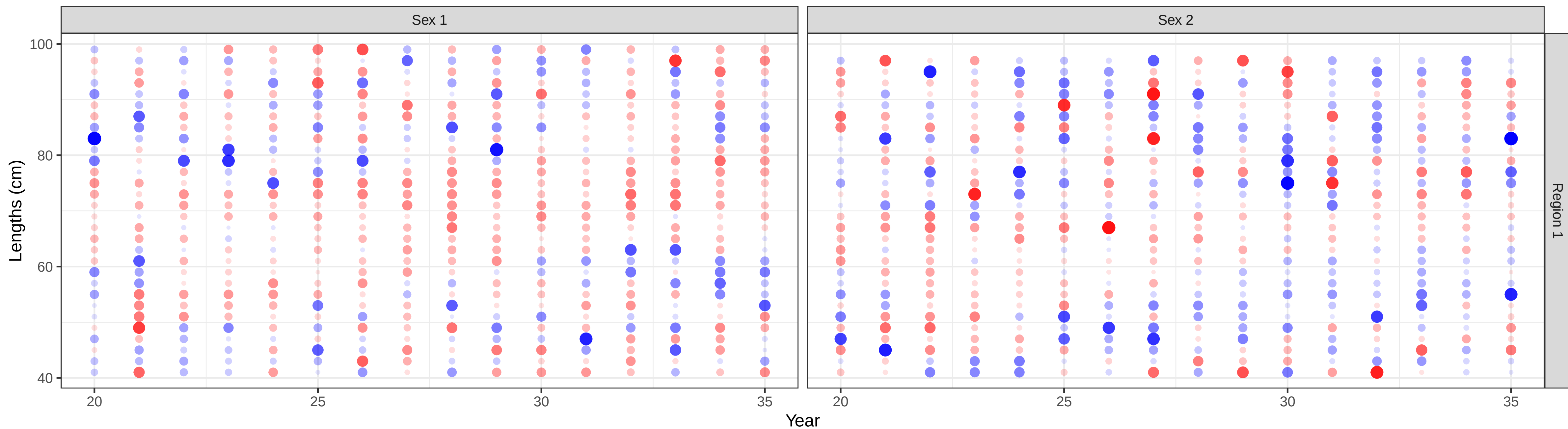




abs(Resid) 1 2 3 Sign ● Neg ● Pos

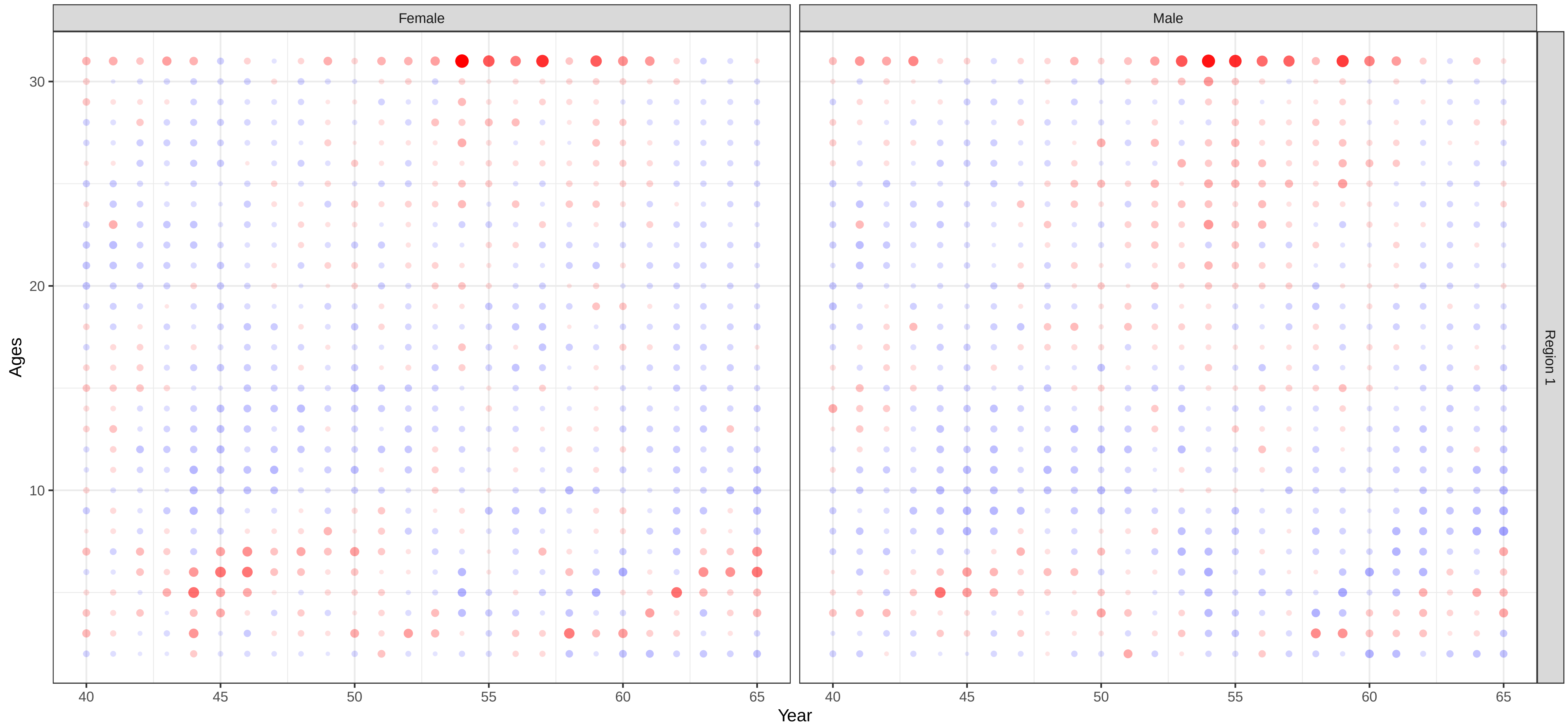


Sign ● Neg ● Pos abs(Resid) ● 1 ● 2



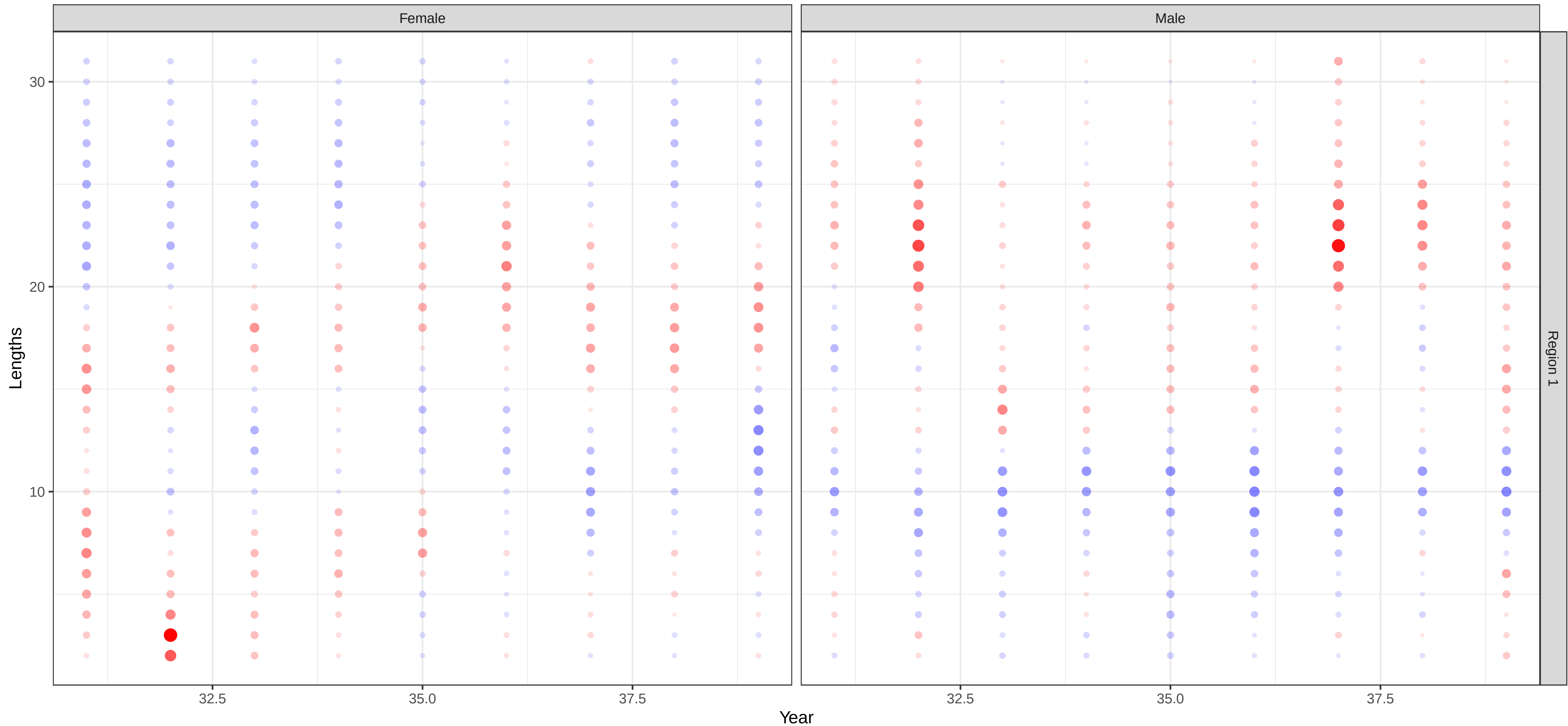
Fixed Gear Fishery Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0



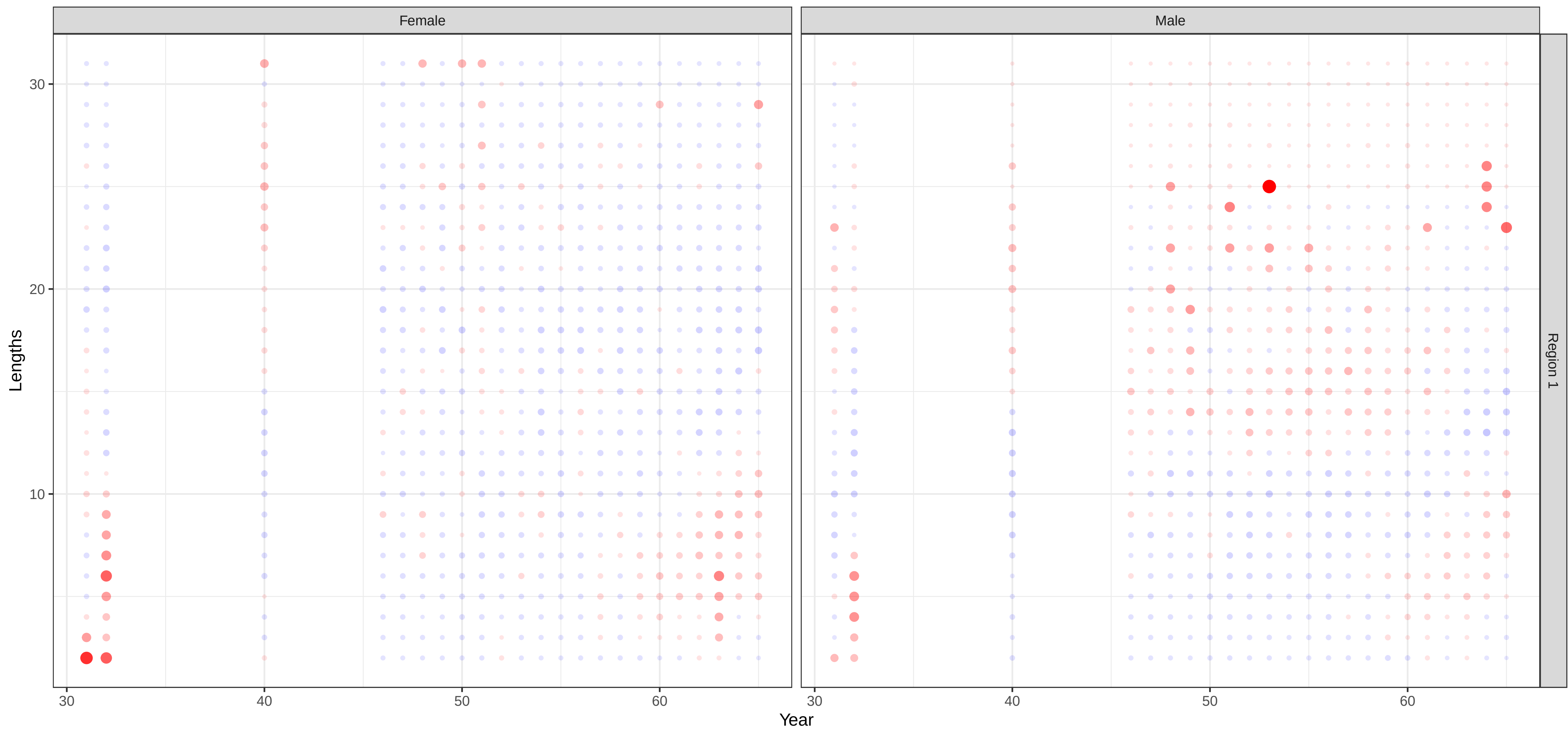
Fixed Gear Fishery Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.0 ● 0.5 ● 1.0 ● 1.5 ● 2.0



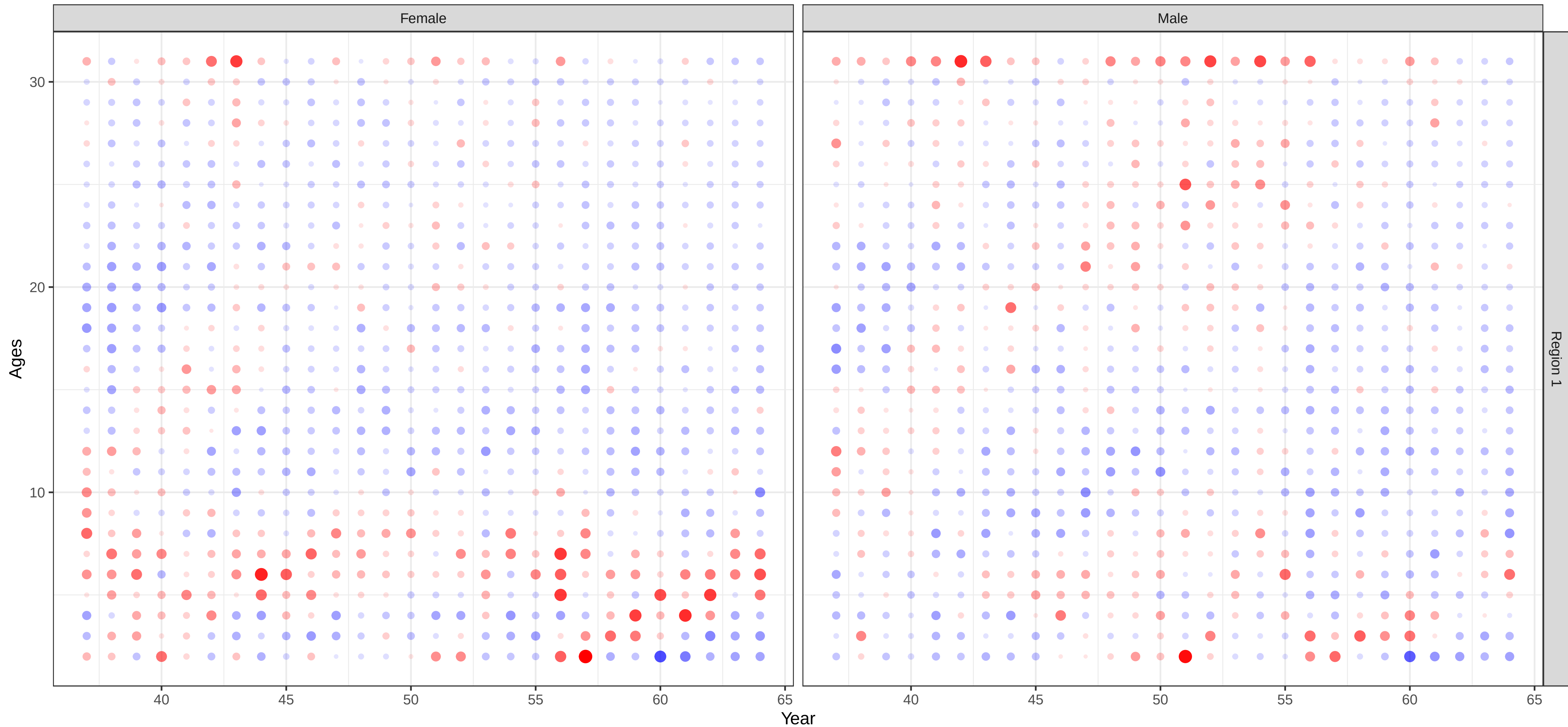
Trawl Fishery Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0 ● 1 ● 2 ● 3 ● 4



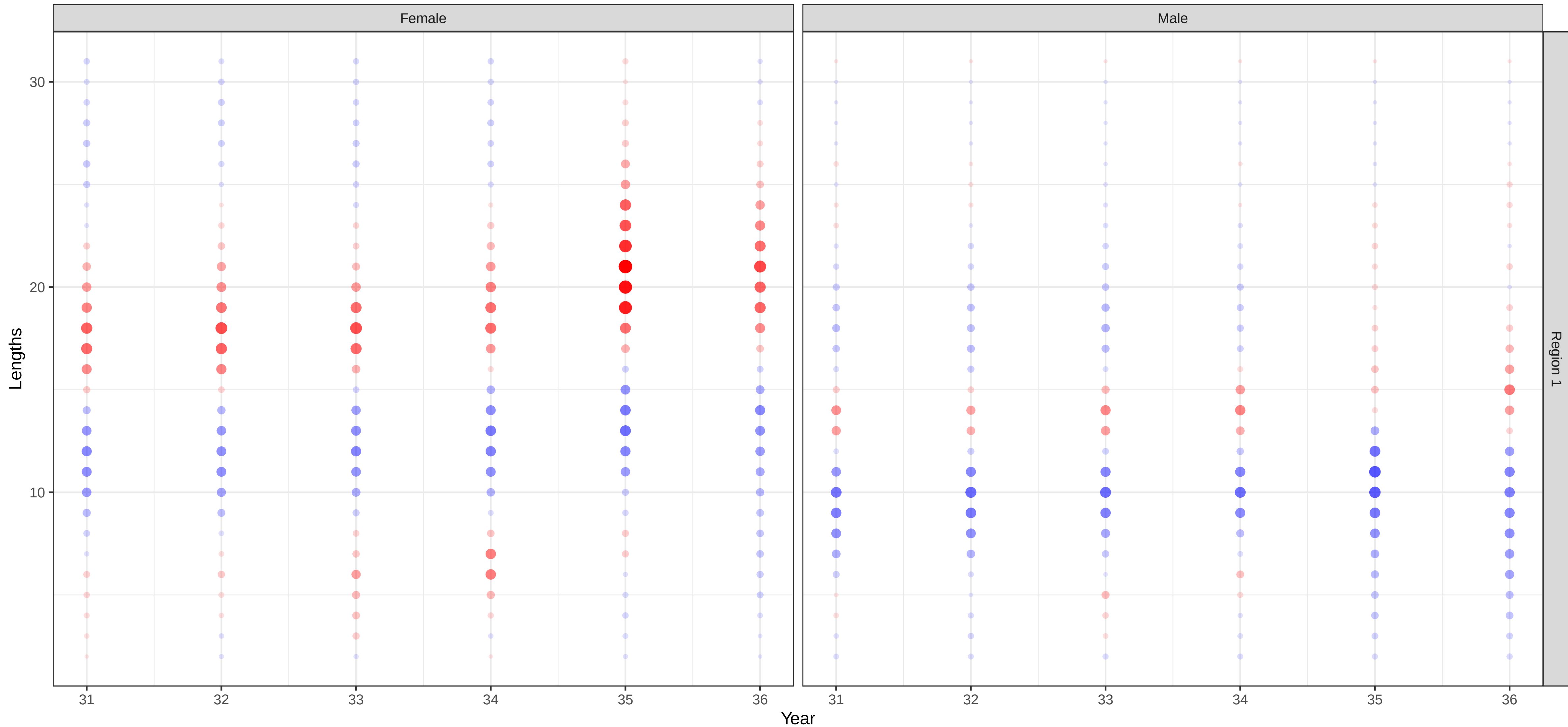
LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0 ● 2.5



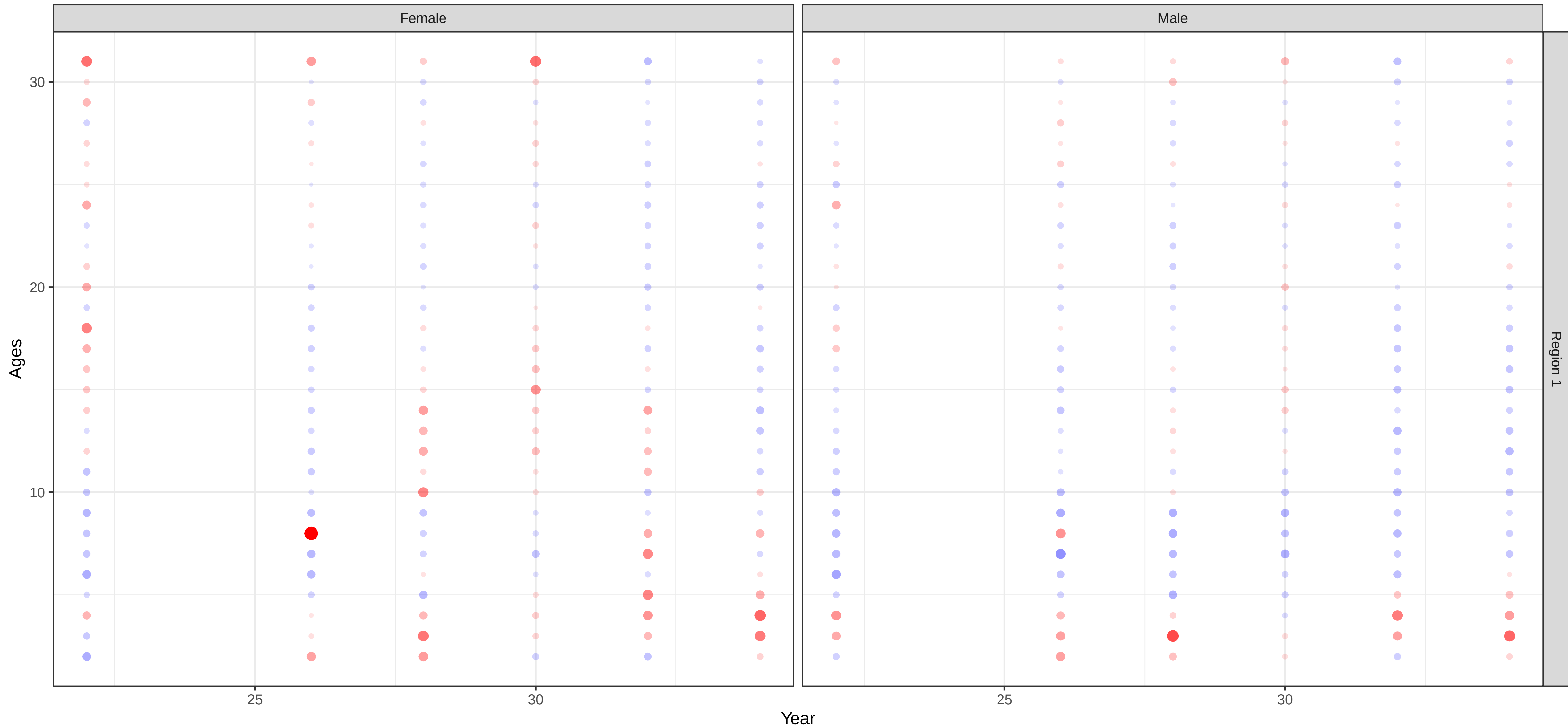
LLS Length Compositions

abs(Resid) 0.0 0.4 0.8 1.2 1.6 Sign ● Neg ● Pos



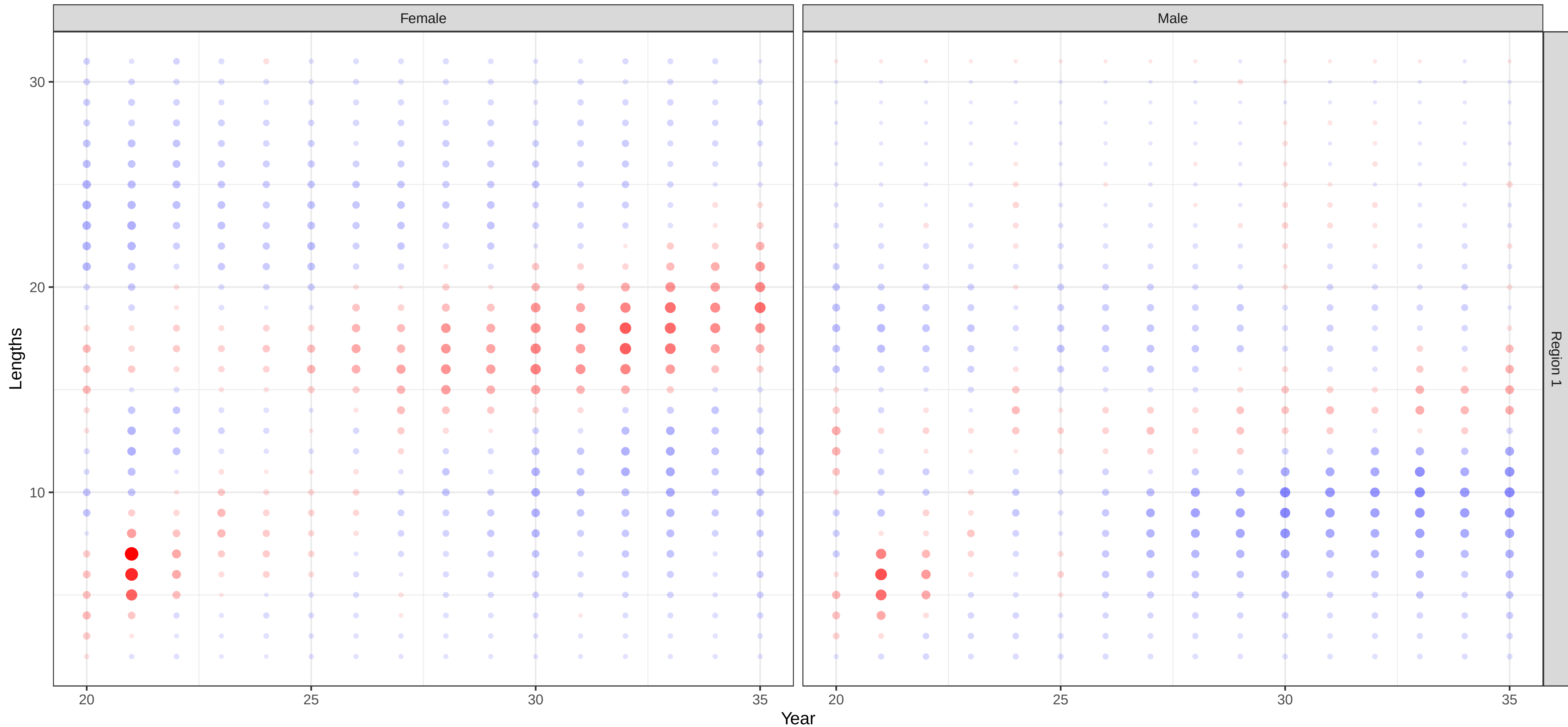
JPN LLS Age Compositions

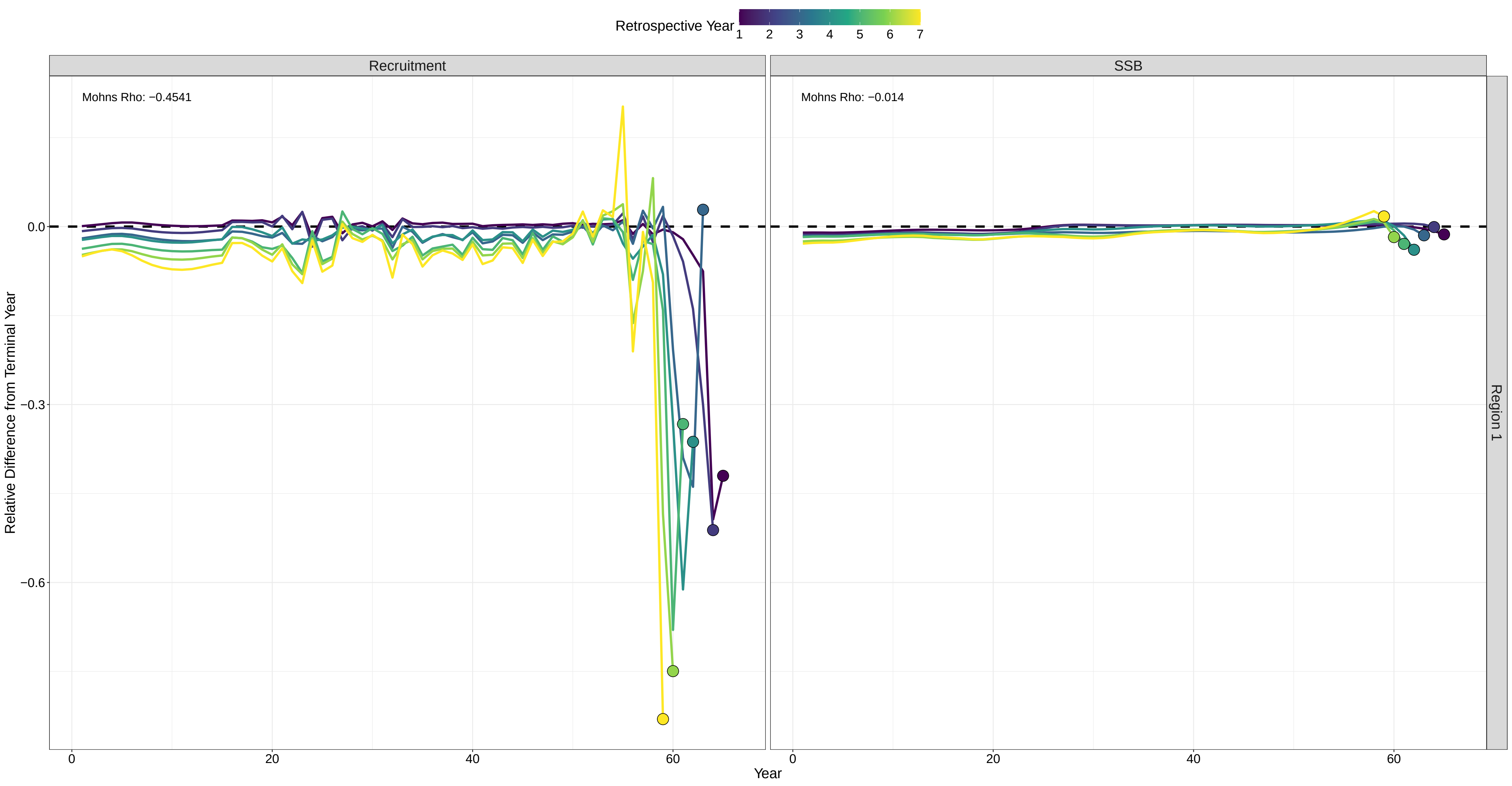
abs(Resid) ● 0.5 ● 1.0 ● 1.5 Sign ● Neg ● Pos

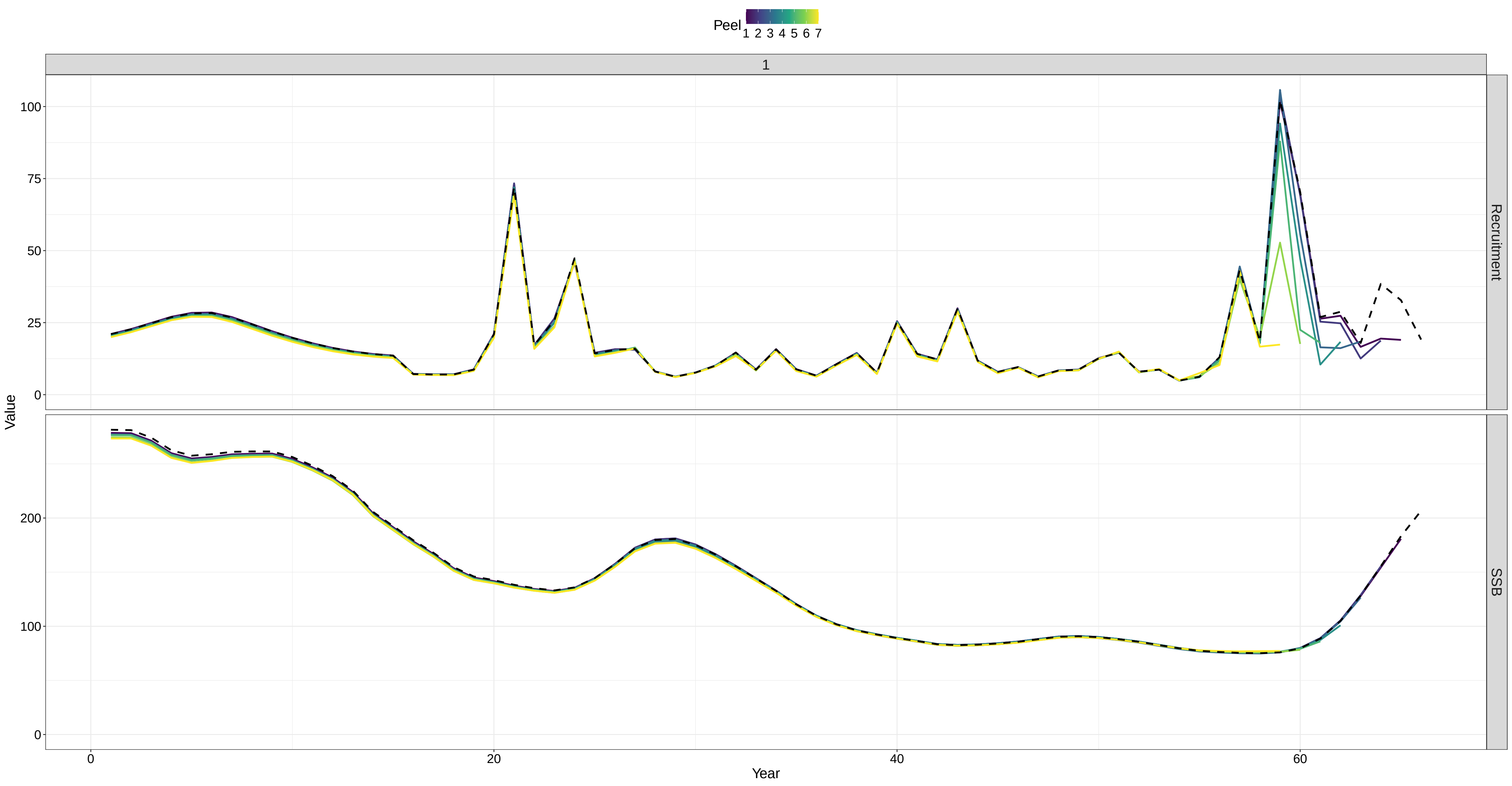


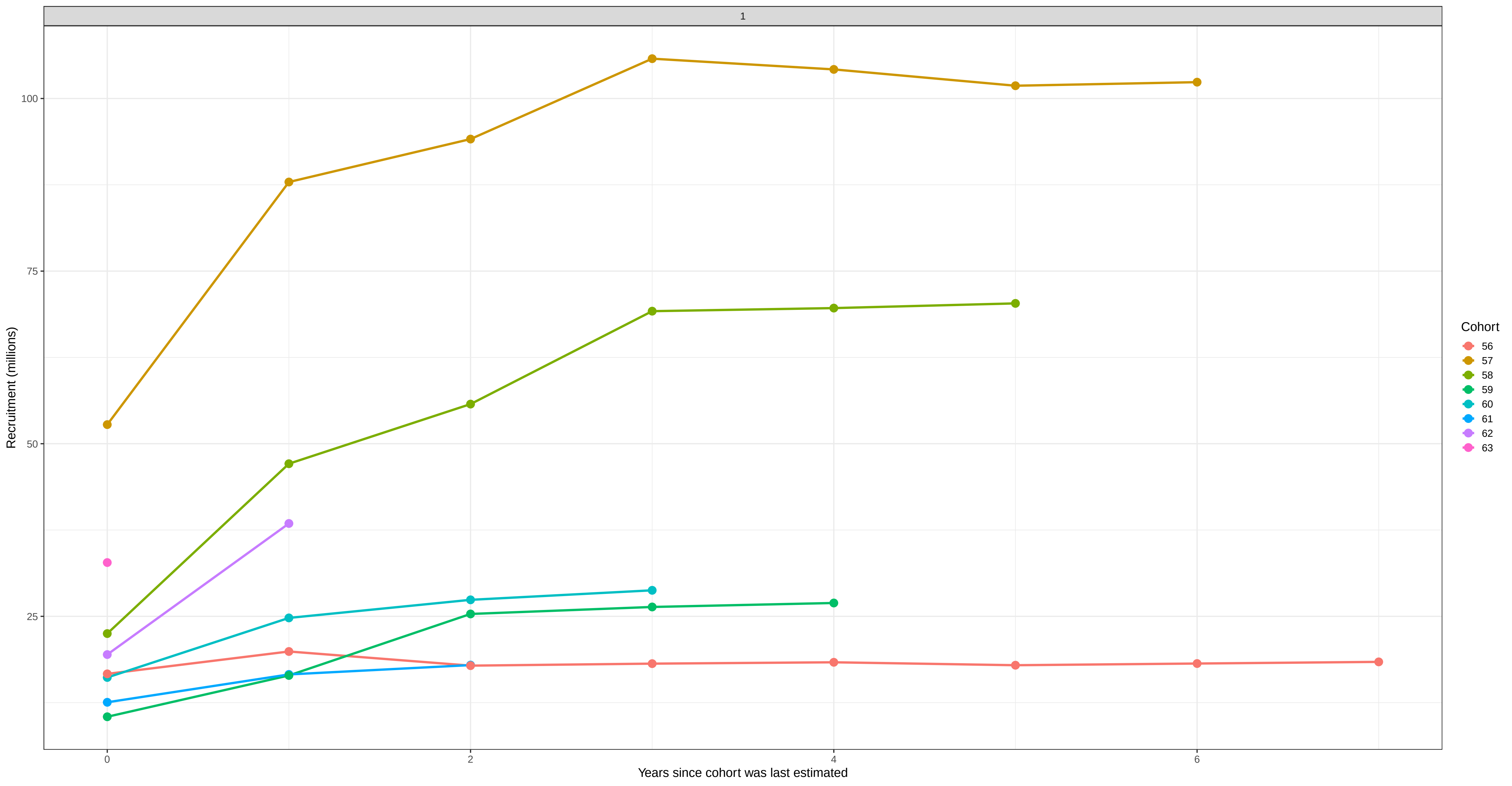
JPN LLS Length Compositions

abs(Resid) 0.0 0.4 0.8 1.2 1.6 Sign Neg Pos

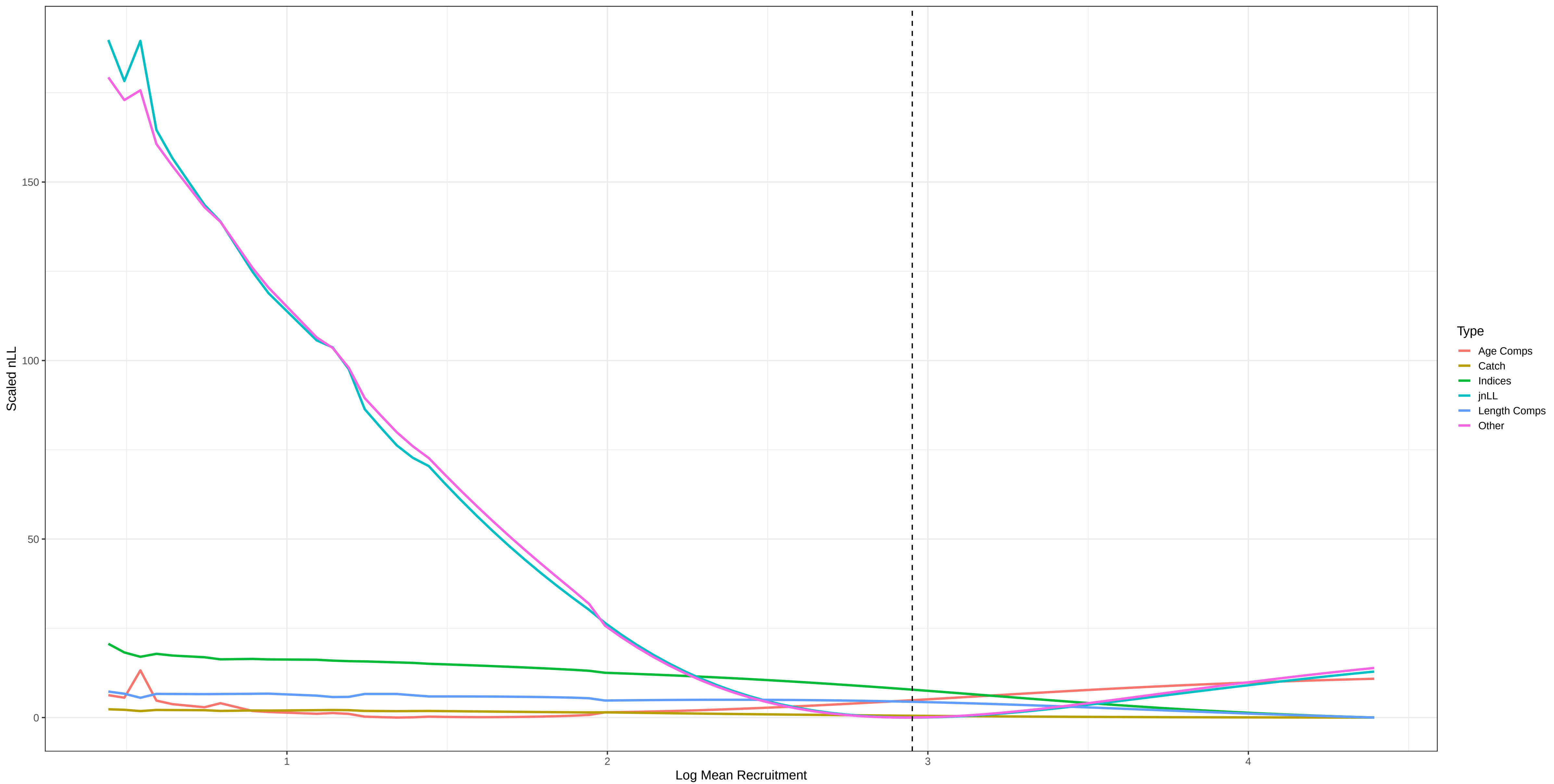




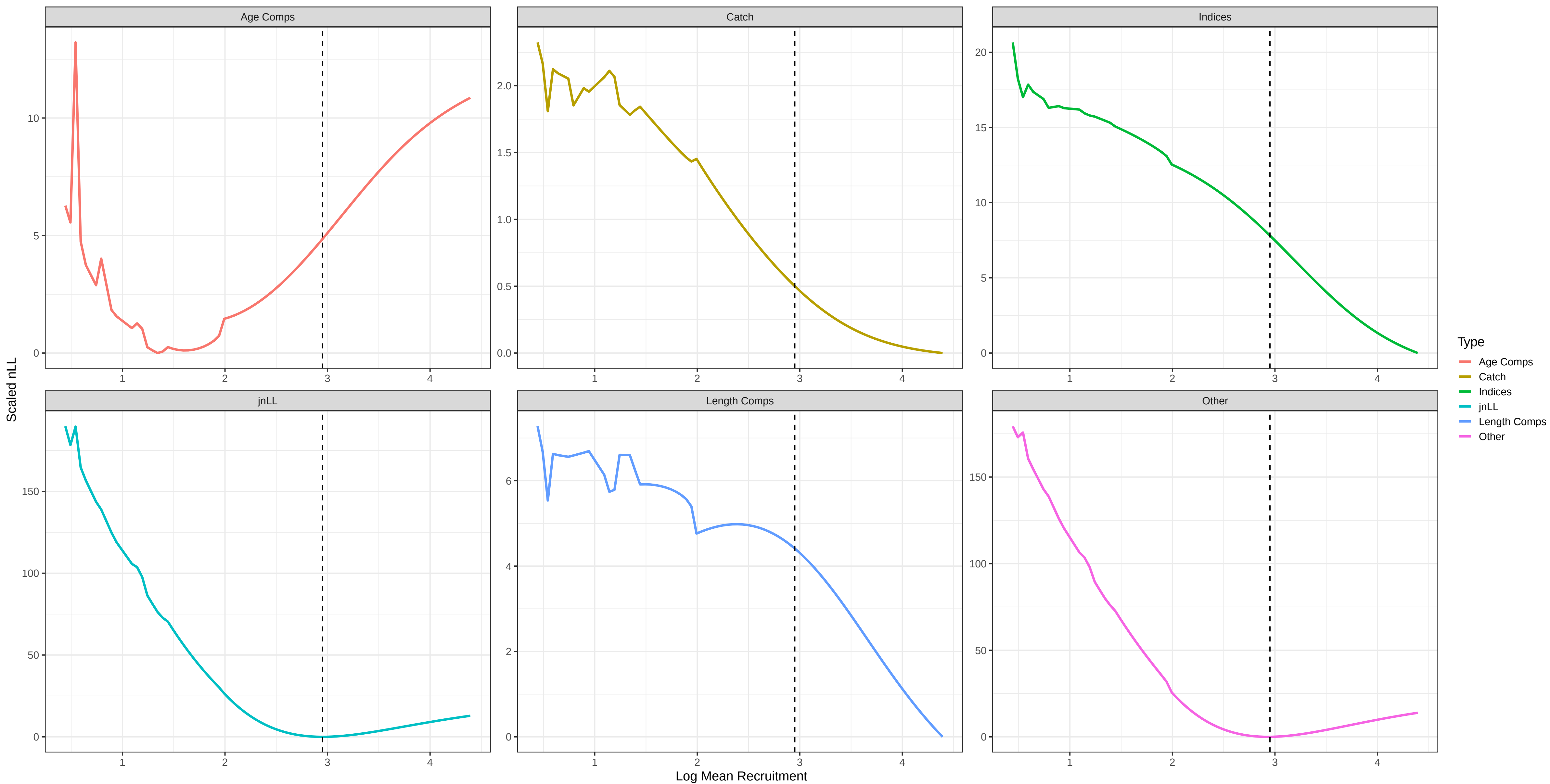




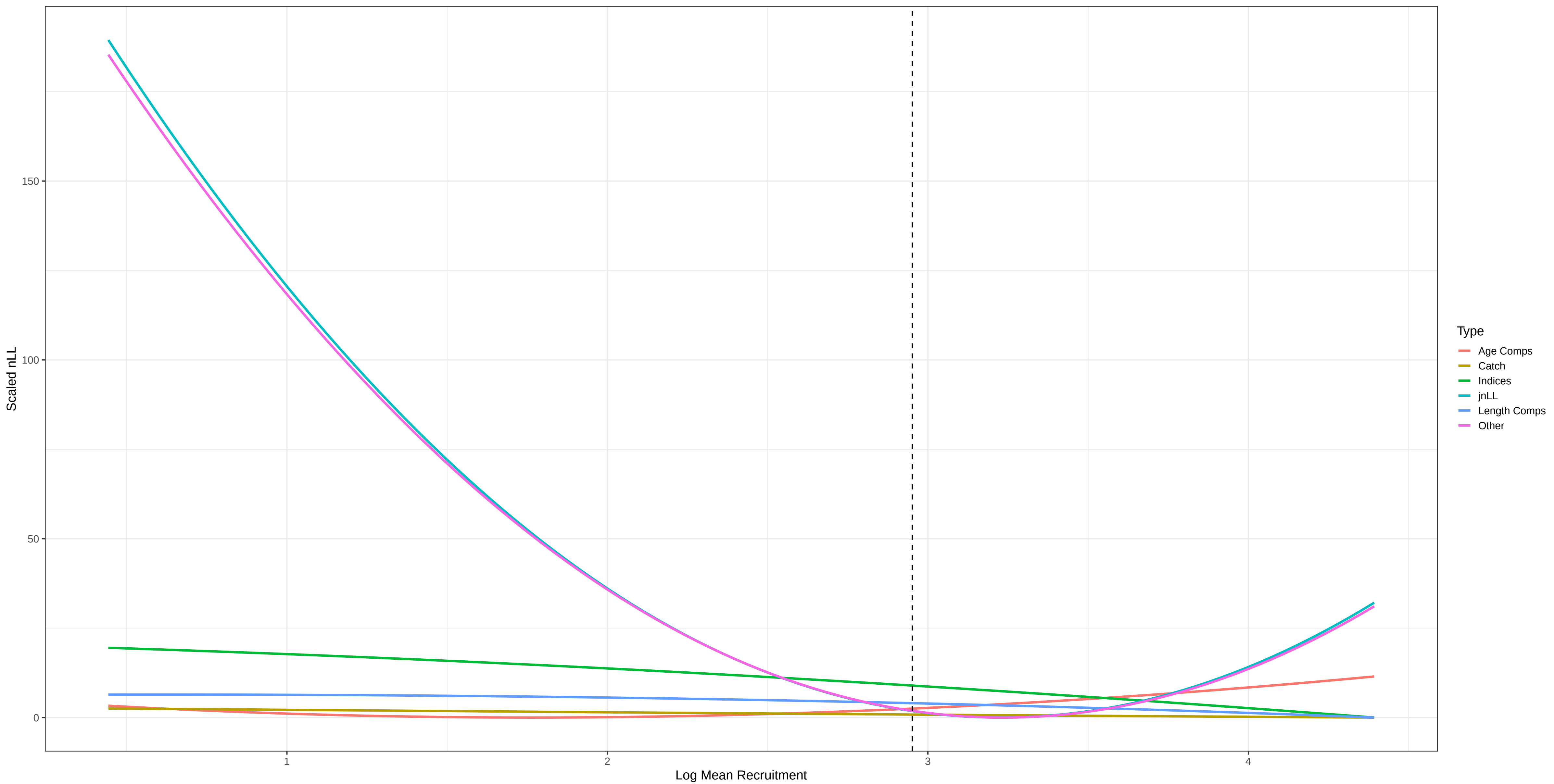
Profile Likelihood R0



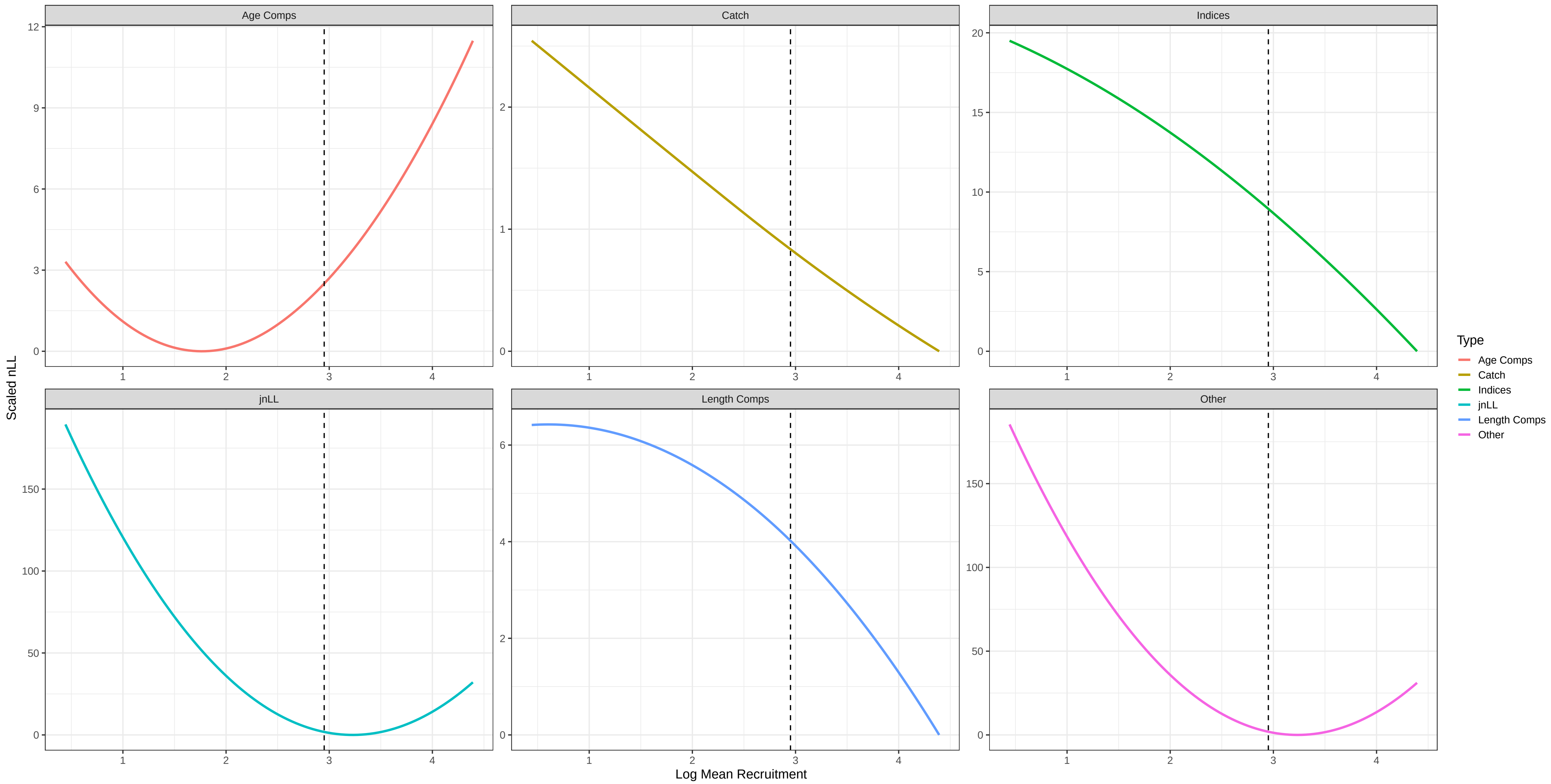
Profile Likelihood R0



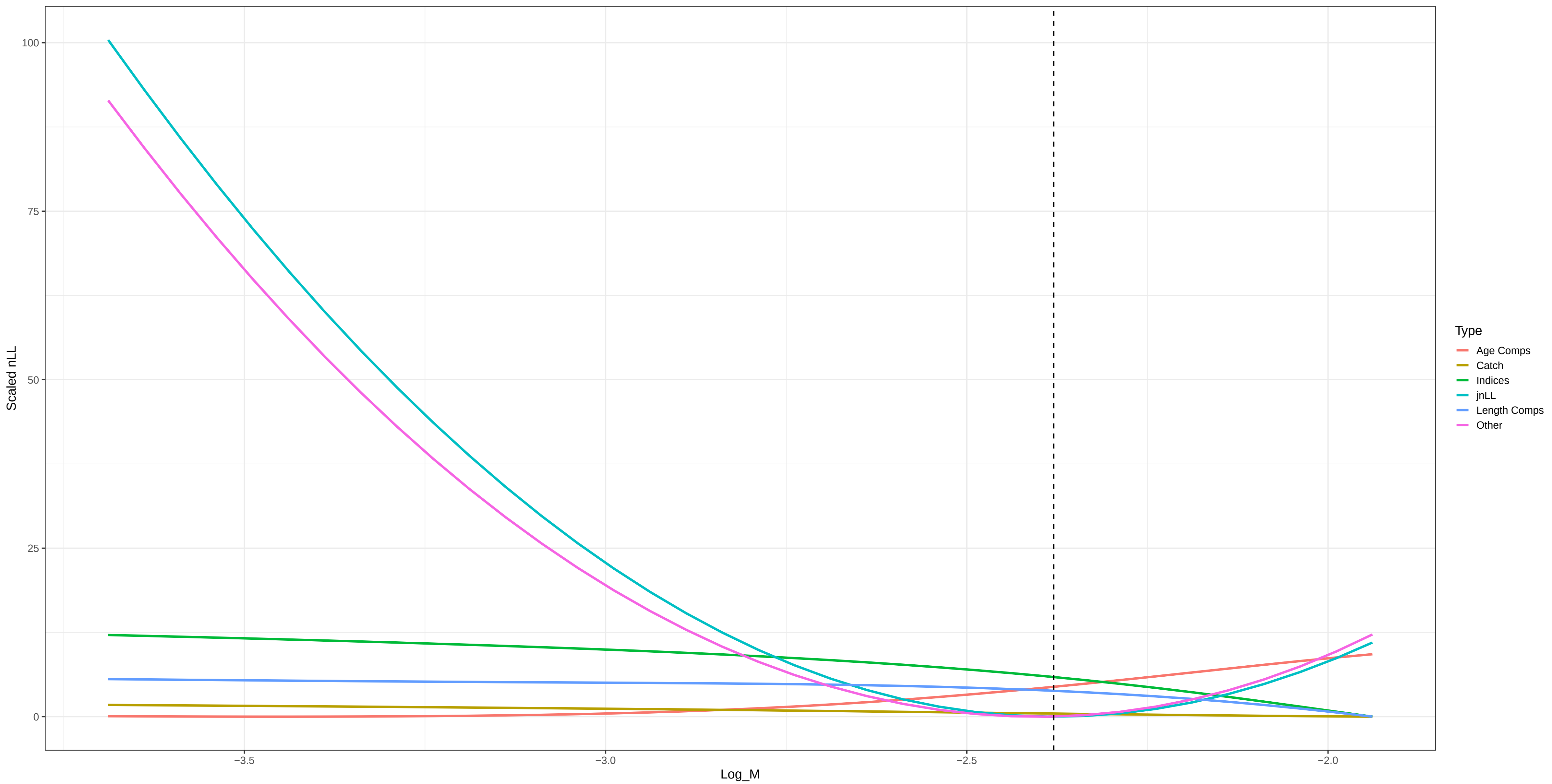
Profile Likelihood R0



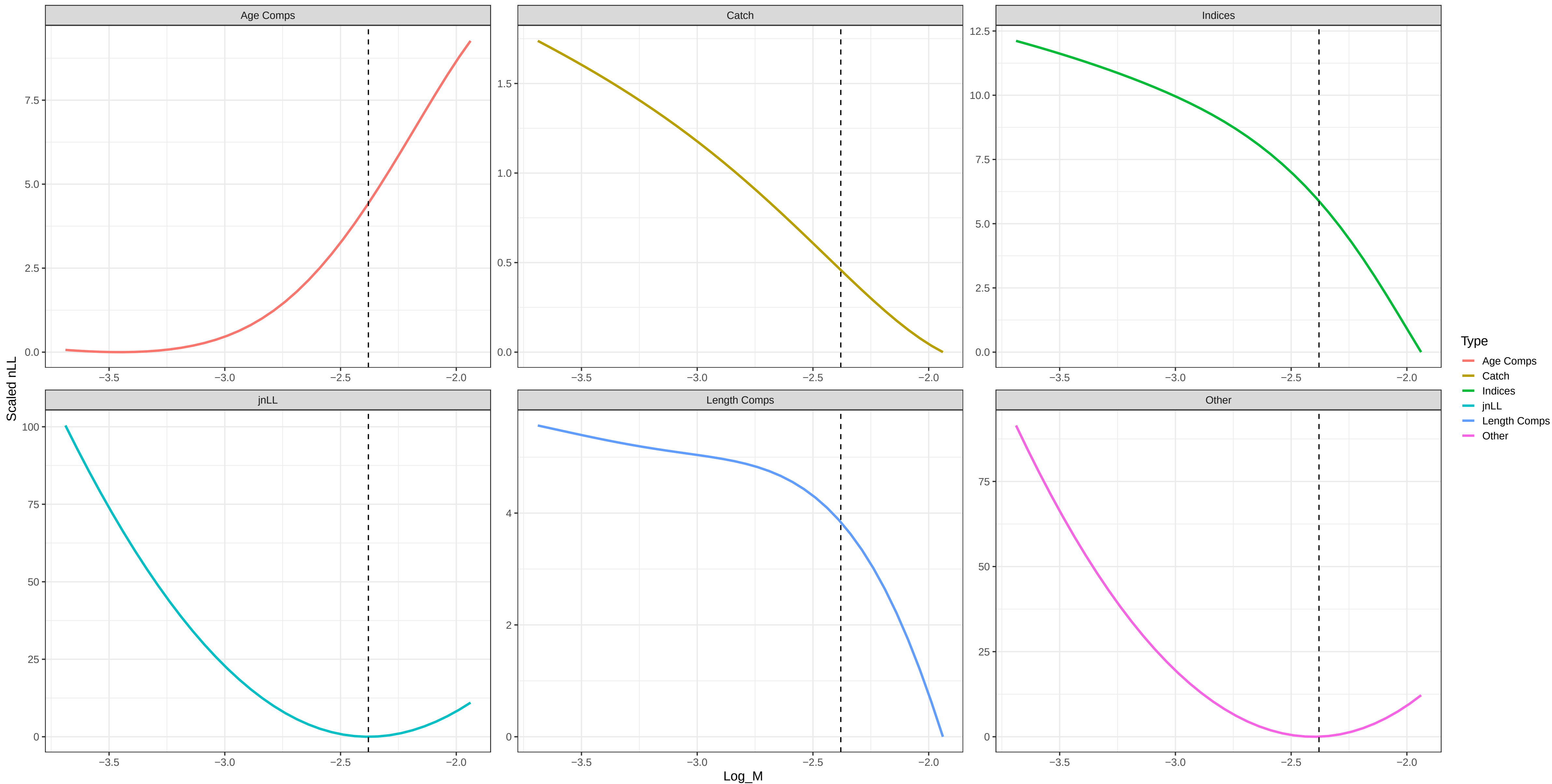
Profile Likelihood R0



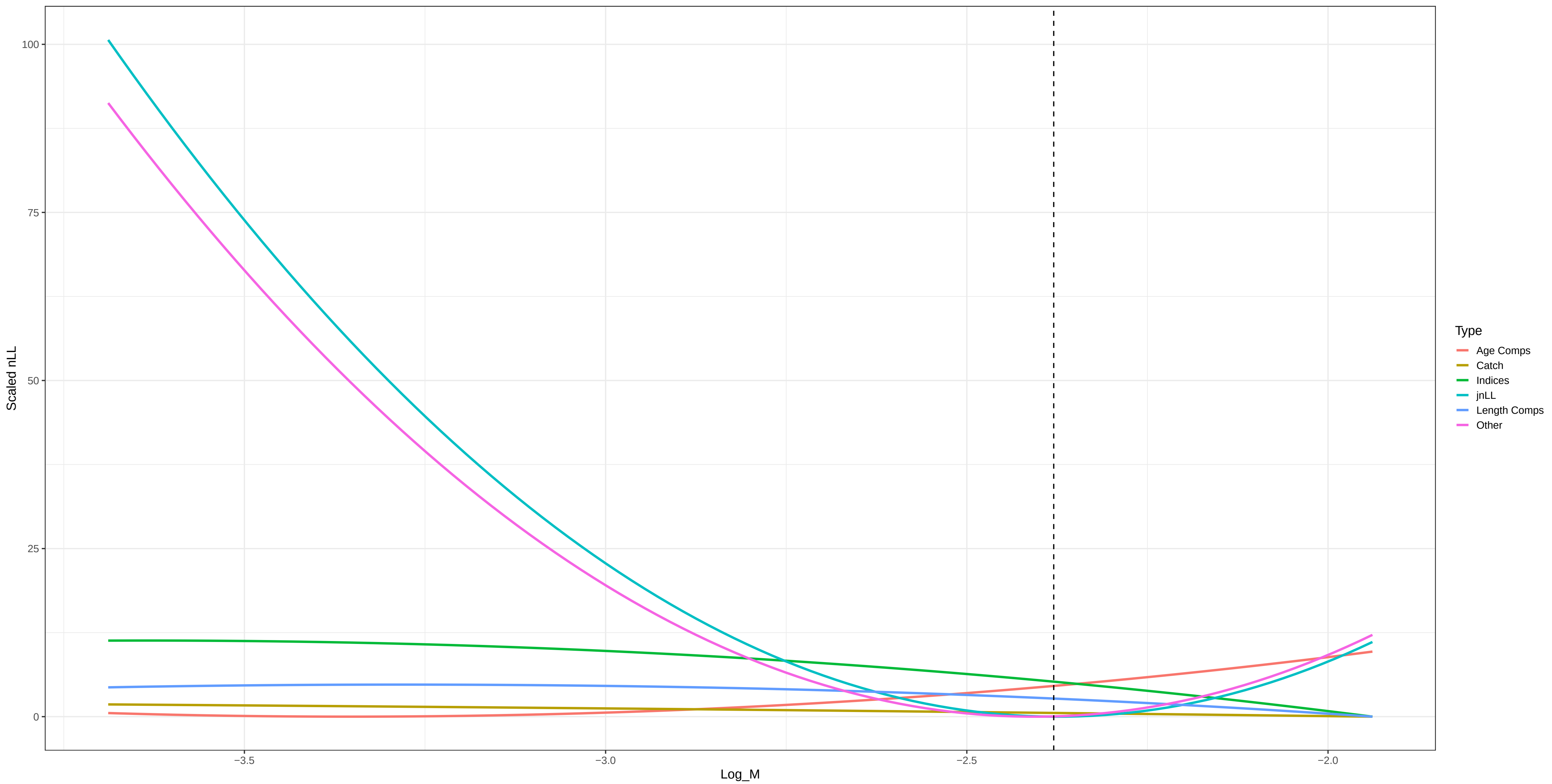
Profile Likelihood M



Profile Likelihood M



Profile Likelihood M



Profile Likelihood M

